

Dropout Universality: Scaling Laws and Optimal Scheduling at the Edge-of-Chaos

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Abstract

We develop a mean-field theory of dropout as a perturbation of critical signal propagation at the edge of chaos. Dropout shifts the perfect-alignment fixed point, making the depth scale for information propagation finite even at critical initialization. We derive critical and crossover scaling laws for correlation decay and establish that smooth activations and kinked (ReLU-type) activations constitute distinct universality classes, with different critical exponents and a universal two-parameter scaling collapse in detuning and dropout strength. The distinction traces to the analytic structure of the correlation map: smooth activations admit a Taylor expansion near perfect alignment, while kinked activations develop a branch point with universal non-analyticity. As a corollary, the framework yields principled dropout schedules that maximize effective correlation length under fixed budget. We validate the theoretical predictions in MLPs and Vision Transformers, where the predicted schedules outperform constant dropout, illustrating the practical utility of the mean-field approach.

1. Introduction

Mean-field analyses of randomly initialized deep networks reveal an order-to-chaos phase diagram that controls both signal propagation, and gradient penetration with depth (Poole et al., 2016; Schoenholz et al., 2017). In this work, we give a holistic exploration of how this picture is modified by dropout. Dropout is a relevant perturbation: it displaces the critical fixed point and grows under representation-group (coarse-graining) flow (Roberts et al., 2022), pushing the dynamics away from criticality and determining the macroscopic phase. Concretely, dropout deforms

the correlation map (Schoenholz et al., 2017) by adding a correlation-independent shift at perfect alignment, so that perfect correlation between inputs is no longer a fixed point for any nonzero dropout. Thus, even at the edge-of-chaos (Sompolinsky et al., 1988; Packard, 1988; Bertschinger & Natschläger, 2004), the depth correlation length is finite. We interpret this shift as an equation of state for the decorrelation order parameter $m \equiv 1 - c^*$ (with c^* the asymptotic inter-signal correlation), derive the resulting scaling laws, and show that smooth and kinked (ReLU-type) activations exhibit the same qualitative phase diagram but different critical scaling in the presence of dropout.

Additionally, we show that the two control parameters (dropout strength and distance to criticality) admit a scaling collapse into a single universal form, and we illustrate how these predictions compare with mean-field recursions (and near-mean-field MLP experiments) in the regimes where finite-width effects remain subdominant.

This analysis yields several theoretical contributions: (i) we prove that smooth and kinked activations belong to distinct universality classes with quantitatively different critical exponents (Table 2); (ii) we derive the Landau equation of state for the decorrelation order parameter and establish a two-parameter scaling collapse; (iii) we show that the dropout field h is a more relevant perturbation than thermal detuning t , with scaling dimension $y_h = 2 > y_t = 1$ for smooth activations; and (iv) we connect the universality class distinction to the Hermite spectral structure of activations (App. F). As a corollary, we derive optimal dropout schedules from first principles and validate these predictions experimentally.

Our work here extends previous literature where variable depth (Huang et al., 2016) was studied by dropping whole residual blocks with increasing probability, curriculum dropout scheduling, where there is dropout annealing over time (Morerio et al., 2017), or LayerDrop (Fan et al., 2019), where whole layers in transformers are dropped for efficiency. Our methods are orthogonal to these as they can be applied simultaneously. Additionally, a distinction in the topological nature of the critical surface in kinked and smooth activation functions was previously explored in (Hayou et al., 2019), while here we extend this by extract-

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

ing the scaling exponents to show that they live in distinct universality classes.

2. Mean-field theory background

Here we briefly review the mean-field assumptions and extent to which mean-field theory (MFT) can be applied in the context of MLPs (Poole et al., 2016; Schoenholz et al., 2017), from which we will build upon. We consider fully-connected MLPs at random initialization, where preactivations are well-approximated as Gaussian random variables with self-consistently determined covariances. In the strict $N_l \rightarrow \infty$ limit this Gaussian description closes, while at large but finite width deviations from Gaussianity can be organized perturbatively in the depth-to-width aspect ratio (schematically L/N), as emphasized in (Roberts et al., 2022).

We consider a MLP of depth L , N_l Neurons in layer l , and an activation $\phi : \mathbb{R} \rightarrow \mathbb{R}$ (e.g. tanh, ReLU, etc.). We use the standard i.i.d. Gaussian initialization

$$W_{ij}^l \sim \mathcal{N}\left(0, \frac{\sigma_w^2}{N_l}\right), \quad b_i^l \sim \mathcal{N}(0, \sigma_b^2), \quad (1)$$

take the input as $y_{i;a}^0 = x_{i;a}$, and propagate forward via

$$z_{i;a}^l = W_{ij}^l y_{j;a}^l + b_i^l, \quad y_{i;a}^{l+1} = \phi(z_{i;a}^l), \quad (2)$$

where a labels the input. Throughout, expectations $\mathbb{E}[\cdot]$ are over the random weights and biases at initialization, with inputs held fixed. We also use the standard Gaussian measure

$$\int Dz(\cdots) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-z^2/2}(\cdots),$$

and similarly for $\int Dz_1 Dz_2$.

In the infinite-width limit this becomes exact layer-by-layer, while at finite width the first layer is exactly Gaussian (being a linear map of fixed inputs) and subsequent layers are only approximately Gaussian, with controlled non-Gaussian corrections parametrized by the depth-to-width ratio (Roberts et al., 2022). A more in-depth treatment of the background machinery needed of MFT is presented in App. B.

We track the layer-wise preactivation variance for a single input q_{aa}^l , and the two input correlation c_{ab}^l where the

covariance of two signals at layer l q_{ab}^l is defined as:

$$q_{aa}^l = \sigma_w^2 \int Dz \phi^2\left(\sqrt{q_{aa}^{l-1}} z\right) + \sigma_b^2 \quad (3)$$

$$q_{ab}^l = \sigma_w^2 \int Dz_1 Dz_2 \phi(u_1) \phi(u_2) + \sigma_b^2 \quad (4)$$

$$u_1 = \sqrt{q_{aa}^{l-1}} z_1, \quad c_{ab}^{l-1} = \frac{q_{ab}^{l-1}}{\sqrt{q_{aa}^{l-1} q_{bb}^{l-1}}}, \quad (5)$$

$$u_2 = \sqrt{q_{bb}^{l-1}} \left(c_{ab}^{l-1} z_1 + \sqrt{1 - (c_{ab}^{l-1})^2} z_2 \right). \quad (6)$$

In the absence of dropout, for a wide array of activations q_{ab}^l, c_{ab}^l settle to fixed points q^*, c^* , in particular, there exists a $c^* = 1$ fixed point. Physically, the variance fixed point is the average variance of activations, while the correlation fixed point tells us that all signals are driven to an indistinguishability due to the stochastic noise. We will denote the recurrence relation for c_{ab}^l as $c_{ab}^l = F(c_{ab}^{l-1})$.

To probe its stability, one linearizes the map at $c = 1$, defining the (angular) susceptibility

$$\chi_{\perp} \equiv \left. \frac{\partial c_{ab}^l}{\partial c_{ab}^{l-1}} \right|_{c_{ab}=1} = \sigma_w^2 \int Dz [\phi'(\sqrt{q^*} z)]^2. \quad (7)$$

This creates a phase diagram with three regimes: $\chi_{\perp} < 1$ is the ordered regime, $\chi_{\perp} > 1$ is chaotic, and $\chi_{\perp} = 1$ defines the critical regime (the so called "edge-of-chaos"). The depth scale controlling cross-input correlations can become arbitrarily large in the latter, allowing information to penetrate deep into the network. As an example of its relevance, He (He et al., 2015) initialization can be derived by requiring criticality for ReLU initialization (see App. B), a more fundamental viewpoint than variance preservation.

This leads to an emergent characteristic depth scale ξ_c , dual-purpose metric that not only parametrizes signal propagation, but also gradient flow in backpropagation. Iterating random affine maps induces an effective coarse-graining in depth akin to the renormalization group flow in theoretical physics, coined representation group flow in the ML context (Roberts et al., 2022); information about norms and angles relaxes exponentially fast

$$|c^* - c_{ab}^l| \propto e^{-l/\xi_c}, \quad \xi_c^{-1} \equiv -\log |r_c|. \quad (8)$$

With an equivalent correlation length defined for q^* which controls how quickly single-input norms settle to q^* . Heuristically, ξ_c controls how far distinctions between different inputs survive with depth; in the ordered phase $c^* = 1$, so $u_1^* = u_2^*$ and $\xi_c^{-1} = -\log \chi_{\perp}$, hence $\chi_{\perp} \rightarrow 1$ implies $\xi_c \rightarrow \infty$. Consequently, the edge-of-chaos is primarily about ξ_c rather than ξ_q , except in special cases. For ReLU,

the curvature vanishes everywhere away from the origin, leading to a coincidence between ξ_c and ξ_q , both diverging as $\chi_\perp \rightarrow 1$, provided a finite q^* exists. This is one concrete reason why ReLU-style near-critical initializations are comparatively forgiving in practice.

Mean field provides a good approximation in very wide networks, especially through the first few layers. At large but finite width, deviations from Gaussian mean-field behavior can be organized perturbatively as an expansion in depth-to-width ratio L/N . Accordingly, in our experiments we work in a controlled regime with $L/N \ll 1$ (i.e., $N \gg L$), which is also the regime in which networks perform best empirically (Roberts et al., 2022). The correlation length diverges as we approach criticality, amplifying finite-width and other non-Gaussian corrections, hence, when probing critical power laws while maintaining dropout, we keep the dropout rate parametrically small but non-negligible: taking dropout to be infinitesimal would require prohibitively deep networks to obtain precise estimates of the correlation length and related quantities, making the experiments computationally unfeasible.

Although we derive these results for MLPs, analogous large-width limits exist for other architectures. For CNNs, near-Gaussian behavior can arise by taking the number of channels to infinity (LeCun et al., 1998; Xiao et al., 2018). For transformers, related infinite-width analyses can be formulated in limits involving large hidden dimension and/or large numbers of attention heads (Vaswani et al., 2017; Hron et al., 2020; Halverson, 2024). We emphasize that the scaling laws we derive are controlled primarily by the analytic structure of the activation function rather than by a particular architecture, and we therefore expect them to be broadly applicable. In section 3.3, we validate our claims in transformers, and show that the intuitions about signal propagation developed in MLPs do in fact translate to transformers.

In App. A we probe different realistic scenarios across architectures, datasets, and schedules. We work in the common regime where our network is in overparameterized settings and dropout aids generalization and accuracy, and we will explicitly explore the trade-off between dropout-driven regularization and preserving stable signal propagation. We show that accumulating dropout in the first layers works best, as the coadaptation tends to concentrate earlier in the architecture, thus improving the regularization, and effective propagation length, explicitly showing how these schedules improve current baselines.

3. Universality Classes and Critical Scaling under Dropout

Dropout (Srivastava et al., 2014) is often used as a regularizer that reduces overfitting by randomly masking ac-

tivations during training, thereby reducing the network’s reliance on any single subset of pathways and discouraging co-adaptation. In the previous literature, dropout was explored in MFT (Lee et al., 2019; Schoenholz et al., 2017), but not in the universality class sense. For the purpose of clarity, we will be completely explicit about what we mean by dropout. We model it as multiplicative Bernoulli noise applied to the activations,

$$p_j^{(\ell)} \sim \text{Bernoulli}(\rho), \quad y_j^{(\ell)} \mapsto \tilde{y}_j^{(\ell)} \equiv \frac{p_j^{(\ell)}}{\rho} y_j^{(\ell)},$$

i.e. the standard inverted-dropout convention in which $\mathbb{E}_p[\tilde{y}_j^{(\ell)}] = y_j^{(\ell)}$, and hence $\mathbb{E}_p[\tilde{z}_i^\ell | W, b] = z_i^\ell$, so the mean preactivation is invariant to the dropout rate (Schoenholz et al., 2017). Throughout, ρ is the keep probability. When comparing two inputs, we draw the dropout masks independently for each input¹; this independence is the source of the decorrelation effect analyzed below.

In this section, we motivate and demonstrate that smooth and kinked (non-smooth) activation functions exhibit demonstrably different behavior when analyzing the asymptotic scaling of mean-field observables near the edge-of-chaos, including the cutoff of the depth correlation length induced by dropout. This distinction was motivated by a variational problem for the correlation length: in the near-critical Gaussian channel, Hermite polynomials form a natural basis, and common activations such as tanh or ReLU admit a Hermite decomposition whose coefficients control how strongly the activation perturbs long-range correlation propagation. The smoking gun is that kinks produce algebraic (power-law) tails in the Hermite spectrum, spreading weight across many degrees and thus many effective scales, whereas smooth (more precisely, analytic) activations have rapidly decaying coefficients, with spectral weight concentrated in low-order modes and an essentially exponential falloff. The details of these computations can be found in App. F.

Our classification is orthogonal to the scale-invariant vs. non-scale-invariant distinction of (Roberts et al., 2022), the multi-scale Hermite spectrum highlighted above is controlled primarily by analytic regularity (kinks and other non-analytic structure), not by scale invariance per se. What we propose here is based on (a) the near-critical scaling laws differ between smooth and kinked activations, and (b) the Hermite spectral weight of non-analytic activations decays as a power law, in contrast to the exponential decay

¹The detuning can be alleviated via correlated dropout masks; while independent dropout masks across datapoints inject sample specific multiplicative noise (Bishop, 1995) that destroys the $c = 1$ fixed point. Instead, if the same mask is shared across the batch, the fixed point structure is preserved and near critical scaling can be recovered by retuning, but the noise becomes batch coupled and the regularization is correspondingly weaker and more structured. We leave detailed analysis of correlated masks to future work.

characteristic of smooth activations.

It is in this context that we seek a sharper analytical understanding of how dropout deforms the edge-of-chaos. Extending mean-field propagation to include dropout, one finds that for a single input dropout largely acts as a renormalization of the effective weight variance, $\sigma_w^2 \mapsto \sigma_w^2/\rho$, but for two inputs the perfect-alignment fixed point at $c^* = 1$ is lost for any $\rho < 1$ when the two dropout masks are chosen independently (Schoenholz et al., 2017). In other words, dropout acts as an explicit symmetry-breaking “field” for alignment: even if two inputs are perfectly aligned at some depth, the next layer produces $\bar{c}_{ab} < 1$, cutting off the divergence of the depth correlation length at the edge.

To begin, we drop activations at random with keep probability ρ , so the preactivations are

$$z_i^\ell = \frac{1}{\rho} \sum_j W_{ij}^\ell p_j^\ell y_j^\ell + b_i^\ell. \quad (9)$$

The single-input variance recursion becomes

$$\bar{q}_{aa}^\ell = \frac{\sigma_w^2}{\rho} \int Dz \phi^2\left(\sqrt{\bar{q}_{aa}^{\ell-1}} z\right) + \sigma_b^2, \quad (10)$$

where the bar reminds us that the variance fixed point is shifted by dropout. We denote the (putative) dropout variance fixed point by

$$\bar{q}^* = \frac{\sigma_w^2}{\rho} \int Dz \phi^2(\sqrt{\bar{q}^*} z) + \sigma_b^2. \quad (11)$$

For two distinct inputs $a \neq b$, we find a dropout shift at perfect alignment. If at some depth $c_{ab}^\ell = 1$, then at the next layer one finds (Schoenholz et al., 2017)

$$\bar{F}_\rho(1) = \bar{c}_{ab}^{\ell+1} = 1 - \frac{1-\rho}{\rho \bar{q}^*} \sigma_w^2 \int Dz \phi^2(\sqrt{\bar{q}^*} z). \quad (12)$$

It is therefore natural to define the dropout field

$$\begin{aligned} h &\equiv 1 - \bar{F}_\rho(1) \\ &= \frac{1-\rho}{\rho \bar{q}^*} \sigma_w^2 \int Dz \phi^2(\sqrt{\bar{q}^*} z). \end{aligned} \quad (13)$$

By construction, $\rho < 1 \Rightarrow h > 0$, so the fixed point at $c^* = 1$ is immediately spoiled: even if two inputs are perfectly aligned at some depth, the next layer produces $\bar{c}_{ab} = 1 - h < 1$. Since we will often be interested in weak dropout, it is useful to record the small- $\delta\rho$ expansion. Writing $\delta\rho \equiv 1 - \rho$ with $\delta\rho \ll 1$, we have $\bar{q}^* = q^* + \mathcal{O}(\delta\rho)$, so to leading order we have

$$h = \delta\rho \frac{\sigma_w^2}{\bar{q}^*} \int Dz \phi^2(\sqrt{\bar{q}^*} z) + \mathcal{O}(\delta\rho^2). \quad (14)$$

3.1. Landau theory and critical exponents

For smooth activations, the dropout-deformed correlation edge admits a simple Landau-style description in terms of three mean-field variables:

$$m \equiv 1 - c^*, \quad t \equiv \chi_\rho - 1, \quad h \equiv 1 - \bar{F}_\rho(1).$$

The fields (m, t, h) are chosen to mirror statistical-mechanics notation: t plays the role of a reduced temperature, m is the order parameter (magnetization), and h is an external field conjugate to m that explicitly biases the system away from perfect alignment.

We assume inverted dropout with keep probability $\rho \in (0, 1]$, with masks drawn i.i.d. across units and independently across inputs. Let $\bar{q}^*$ denote the dropout variance fixed point (11). For a smooth normalizable activation $\phi \in C^2$, repeated application of Price’s theorem gives the first two derivatives of the correlation map at perfect alignment,²

$$\chi_\rho \equiv \bar{F}_\rho'(1) = \sigma_w^2 \int Dz [\phi'(\sqrt{\bar{q}^*} z)]^2, \quad (15)$$

$$g_\rho \equiv \bar{F}_\rho''(1) = \sigma_w^2 \bar{q}^* \int Dz [\phi''(\sqrt{\bar{q}^*} z)]^2. \quad (16)$$

For kinked activations (ReLU-type) $\bar{F}_\rho''(1)$ is not finite and the analytic expansion below is not the correct organizing principle; we treat that universality class separately.

The Landau equation of state follows by expanding $\bar{F}_\rho(c)$ around $c = 1$. Writing $c^* = 1 - m$ with $0 < m \ll 1$ acting as the order parameter,

$$\bar{F}_\rho(1 - m) = \bar{F}_\rho(1) - \chi_\rho m + \frac{g_\rho}{2} m^2 + \mathcal{O}(m^3) \quad (17)$$

$$= (1 - h) - \chi_\rho m + \frac{g_\rho}{2} m^2 + \mathcal{O}(m^3), \quad (18)$$

imposing the fixed-point condition $1 - m = \bar{F}_\rho(1 - m)$ yields

$$h + t m - \frac{g_\rho}{2} m^2 = 0, \quad \Longleftrightarrow \quad h = \frac{g_\rho}{2} m^2 - t m, \quad (19)$$

with $t \equiv \chi_\rho - 1$. The physical (nonnegative) solution is

$$m(t, h) = \frac{t + \sqrt{t^2 + 2g_\rho h}}{g_\rho}. \quad (20)$$

At the correlation edge $t = 0$, dropout displaces the fixed point by

$$m(0, h) = \sqrt{\frac{2h}{g_\rho}}. \quad (21)$$

This square-root power law is our first encounter with a larger family of critical exponents familiar in the physics

²The ρ -dependence is implicit through $\bar{q}^*$. The full computation is given in App. B.1.

Exponent	Mathematical Definition
β	$m \sim t ^\beta \quad (h = 0)$
δ	$m \sim h ^{1/\delta} \quad (t = 0)$
γ	$\chi_M \equiv \frac{\partial m}{\partial h} \Big _{h \rightarrow 0} \sim t ^{-\gamma}$
α	$C \equiv -\frac{\partial^2 f}{\partial t^2} \sim t ^{-\alpha}$
ν_t	$\xi \sim t ^{-\nu_t} \quad (h = 0)$
ν_ρ	$\xi \sim h ^{-\nu_\rho} \quad (t = 0)$
p	$m \sim \ell^{-p} \quad (t = 0, h = 0)$

Table 1. Definitions of critical exponents in the thermodynamic limit ($t \rightarrow 0, h \rightarrow 0$).

literature, defined in Table 1. The key exponents for our analysis are those governing the depth correlation length $\xi_{c,\rho}$, as it sets the characteristic depth scale over which distinct inputs remain distinguishable and therefore provides an effective upper bound on the depth over which the network can reliably propagate information (and learn). Their derivations are straightforward but cumbersome, so we collect them in App. C and simply quote the resulting exponent values in Table 2. For reference, we also list the standard mean-field (Landau) exponents of the Ising universality class (Ising, 1925; Landau & Lifshitz, 1980) and the corresponding mean-field exponents of spin-glass theory (Edwards & Anderson, 1975; Parisi, 1979; 1980; 1983a), since spin-glass models provide one of the closest Statistical Mechanics analogues of neural-network models (Hopfield, 1982; Amit et al., 1985; Amit, 1989), something we will see reflected in the scaling exponents. The exponents quoted in table 2 can be found in (Mézard et al., 1987; Parisi, 1983b; Sherrington & Kirkpatrick, 1975).

Universality Class	ν_t	β	p	γ	δ	ν_ρ	α
Smooth activation	1	1	1	1	2	$\frac{1}{2}$	-1
Kinked activation	1	2	2	1	$\frac{3}{2}$	$\frac{1}{3}$	-3
Spin glass (SK MF)	$\frac{1}{2}$	1	2	1	2	$\frac{1}{4}$	-1
Ising-like	$\frac{1}{2}$	$\frac{1}{2}$	1	1	3	1/3	0

Table 2. Critical exponents for neural network initialization across different universality classes.

3.2. Scaling collapse and crossover region

As we now show, dropout and subcriticality are nothing but two sides of the same coin, and that the equation of state 19 collapses into a function of one variable. This is expected from the point of view of statistical mechanics where universal scaling collapse is a hallmark of criticality. In practice, experiments are rarely tuned exactly to the correlation edge $t = 0$ while simultaneously taking the dropout field to zero, $h \rightarrow 0$. The correct near-critical description is therefore a two-parameter scaling function.

For smooth activations, the order parameter $m \equiv 1 - c^*$ obeys (19),

$$h = \frac{g_\rho}{2} m^2 - t m, \quad t \equiv \chi_\rho - 1. \quad (22)$$

This implies the explicit crossover solution

$$m(t, h) = \frac{t + \sqrt{t^2 + 2g_\rho h}}{g_\rho}. \quad (23)$$

Define rescaled variables

$$\tilde{m} \equiv m \sqrt{\frac{g_\rho}{2h}}, \quad \tilde{t} \equiv \frac{t}{\sqrt{2g_\rho h}}. \quad (24)$$

Then all curves collapse onto the universal scaling function

$$\tilde{m} = \sqrt{1 + \tilde{t}^2} + \tilde{t}. \quad (25)$$

The crossover scale is

$$|t| \sim \sqrt{g_\rho h}. \quad (26)$$

For $|t| \ll \sqrt{g_\rho h}$, one is in the field-dominated regime

$$m \sim \sqrt{\frac{2h}{g_\rho}}. \quad (27)$$

For $|t| \gg \sqrt{g_\rho h}$, the asymptotics depend on the sign of t :

$$t < 0 : \quad m \simeq \frac{h}{|t|}, \quad (28)$$

$$t > 0 : \quad m \simeq \frac{2t}{g_\rho} + \frac{h}{t} \dots \quad (29)$$

Thus, at either side of the crossover window the system is either in the linear-response regime on the subcritical side ($t < 0$), or it sits on the chaotic fixed-point branch on the supercritical side ($t > 0$).

For kinked activations, the leading non-linearity is non-analytic and the equation of state takes the form (see (87)),

$$h = \kappa m^{3/2} - t m + \mathcal{O}(m^2). \quad (30)$$

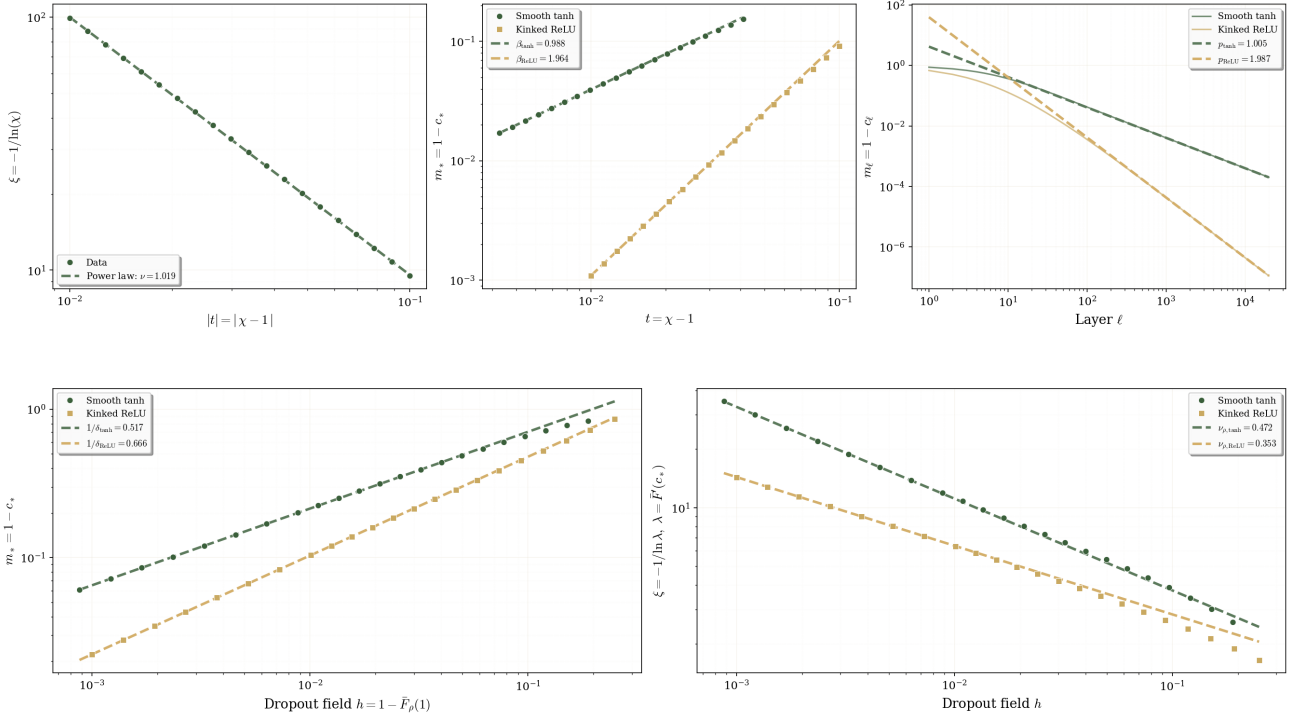


Figure 1. Critical scaling for smooth (tanh) and kinked (ReLU) activation functions, comparing tuning at zero dropout to tuning at the edge-of-chaos using a dropout field. The top row compares the different critical exponents at zero dropout and probes critical detuning decay, while the bottom row explores on critical networks with non-zero dropout. We can appreciate that as the variables grow, the linearized theory diverges from the recursive theory as the higher order ($\mathcal{O}(t^2)$, $\mathcal{O}(m^3)$, $\mathcal{O}(h^2)$...) effects become comparable.

This implies the scaling form

$$m(t, h) = \left(\frac{h}{\kappa}\right)^{2/3} \mathcal{F}\left(\frac{t}{\kappa^{2/3} h^{1/3}}\right). \quad (31)$$

Let

$$u \equiv \frac{t}{\kappa^{2/3} h^{1/3}}, \quad (32)$$

and define $y(u)$ as the positive real root of

$$y^3 - u y^2 - 1 = 0. \implies \mathcal{F}(u) = y(u)^2. \quad (33)$$

The associated crossover scale is

$$|t| \sim \kappa^{2/3} h^{1/3}, \quad (34)$$

showing that kinked activations lie in a distinct universality class from smooth activations.

Practically, this gives us a metric for when the notion of distance to criticality stops being governed by t and starts being governed by h .

3.3. Optimal dropout scheduling and rank-flow tie-breaking

So far we have taken the keep probability to be constant with depth. Since dropout enters as a relevant field that cuts

off the depth correlation length even at the correlation edge, it is natural to ask how to allocate a fixed dropout budget across layers so as to preserve near-critical propagation as much as possible. Let us introduce a depth-dependent keep probability ρ_ℓ and define the associated dropout field

$$h_\ell \equiv \Delta(\rho_\ell), \quad (35)$$

i.e. the shift of the correlation map at perfect alignment induced by keep probability ρ_ℓ . In the smooth universality class, linearizing the correlation map about the dropout-shifted fixed point gives (see (81))

$$\lambda(t, h_\ell) \simeq 1 - \sqrt{t^2 + 2g_\rho h_\ell}. \quad (36)$$

For $0 < \lambda_\ell \lesssim 1$, perturbations propagate multiplicatively,

$$\delta c_L \sim \prod_{\ell=1}^L \lambda_\ell \delta c_0, \quad \lambda_\ell \equiv \lambda(t, h_\ell), \quad (37)$$

which motivates defining an effective inverse correlation length as the mean decay rate per layer,

$$\xi_{\text{eff}}^{-1} \equiv -\frac{1}{L} \sum_{\ell=1}^L \log \lambda_\ell \simeq \frac{1}{L} \sum_{\ell=1}^L \sqrt{t^2 + 2g_\rho h_\ell}. \quad (38)$$

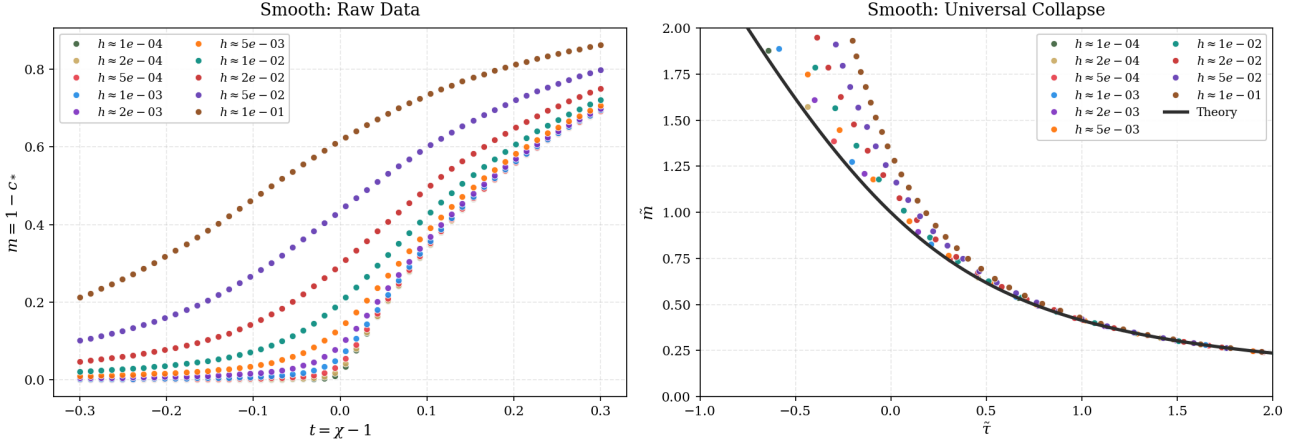


Figure 2. Two-parameter crossover and scaling collapse of the dropout deformed equation of state for smooth universality class (tanh). Plots obtained using MFT recursion relations. The figure displays the relationship between different variables collapsing into a universal function as $\bar{t}$ increases. Kinked counterpart can be found in App. A.

At the correlation edge $t = 0$, this reduces to

$$\xi_{\text{eff}}^{-1}(t = 0) \simeq \sqrt{2g\rho} \frac{1}{L} \sum_{\ell=1}^L h_{\ell}^{1/2} \propto \frac{1}{L} \sum_{\ell=1}^L h_{\ell}^{1/2}. \quad (39)$$

We now fix a total dropout budget and local bounds,

$$\bar{h} \equiv \frac{1}{L} \sum_{\ell=1}^L h_{\ell}, \quad 0 \leq h_{\ell} \leq h_{\max}. \quad (40)$$

Maximizing ξ_{eff} is equivalent to minimizing ξ_{eff}^{-1} , so in the continuum limit ($x = \ell/L$) we obtain the constrained variational problem

$$\min_h \int_0^1 h(x)^{1/2} dx \quad \text{s.t.} \quad \int_0^1 h(x) dx = \bar{h}. \quad (41)$$

Since $h^{1/2}$ is concave, Jensen's inequality implies that any constant schedule maximizes $\int h^{1/2}$ at fixed mean; hence any minimizer must saturate the box constraint and is step-like. Equivalently, any schedule that takes values in $\{0, h_{\max}\}$ with active fraction $f = \bar{h}/h_{\max}$ is optimal. A convenient representative is a step schedule,

$$h(x) = \begin{cases} 0, & x < 1 - f, \\ h_{\max}, & x \geq 1 - f, \end{cases} \quad f = \frac{\bar{h}}{h_{\max}}, \quad (42)$$

for which uniform dropout yields $\xi_{\text{eff}} \sim \bar{h}^{-1/2}$ while the step schedule gives

$$\frac{\xi_{\text{eff,step}}}{\xi_{\text{eff,uniform}}} = \sqrt{\frac{h_{\max}}{\bar{h}}} \geq 1. \quad (43)$$

For kinked activations at $t = 0$, one has $\xi^{-1} \propto h^{1/3}$, so the objective becomes $\int_0^1 h(x)^{1/3} dx$, which is also concave and therefore leads to the same step-like allocation.

At the level of mean-field correlation dynamics, ξ_{eff} depends on $\{h_{\ell}\}$ only through symmetric sums (e.g. (39)), and is therefore invariant under permutations of the depth profile. Mean field fixes the optimal allocation in magnitude (step-like), but cannot by itself select where the saturated block should be placed.

A tie-breaker arises from representation dynamics beyond the mean-field closure. One concrete mechanism is rank collapse. In pure self-attention, for token representations $X^{\ell} \in \mathbb{R}^{n \times d}$ one has (in the absence of residual connections)

$$\text{rank}(X^{\ell+1}) \leq \text{rank}(A^{\ell}) \text{rank}(X^{\ell}), \quad (44)$$

where A^{ℓ} is the attention matrix. Since A^{ℓ} contracts geometrically toward the rank-one projector $\mathbf{1}\mathbf{1}^{\top}/n$, an effective-rank proxy collapses doubly exponentially with depth,

$$R(\ell) \lesssim R(0) \exp(-c2^{\ell}), \quad (45)$$

for an architecture-dependent constant $c > 0$. In wide MLPs, rank proxies typically decay exponentially with depth (e.g. at a rate set by a Lyapunov gap),

$$R(\ell) - 1 \sim (R(0) - 1) \exp(-2(\lambda_1 - \lambda_2)\ell), \quad (46)$$

where $\lambda_1 > \lambda_2$ are the top Lyapunov exponents governing singular-value growth. In both cases, the dominant loss of diversity occurs early in depth, especially in attention-dominated models.

Dropout counteracts this collapse by shrinking within-layer normalized correlations. With keep probability $\rho_{\ell} \in (0, 1]$, inverted dropout acts as

$$\tilde{y}_{\ell,i} = \frac{p_{\ell,i}}{\rho_{\ell}} y_{\ell,i}, \quad p_{\ell,i} \sim \text{Bernoulli}(\rho_{\ell}), \quad (47)$$

with the $p_{\ell,i}$ independent across i . Let the (uncentered) second moment at depth ℓ be

$$\Sigma_\ell \equiv \mathbb{E}[y_\ell y_\ell^\top], \quad (48)$$

and define $\tilde{\Sigma}_\ell$ analogously for $\tilde{y}_\ell$. Averaging over the dropout mask gives the exact second-moment update

$$\tilde{\Sigma}_\ell = \Sigma_\ell + \left(\frac{1}{\rho_\ell} - 1\right) \text{diag}(\Sigma_\ell), \quad (49)$$

so off-diagonal second moments are unchanged while the diagonal is inflated. Writing

$$D_\ell \equiv \text{diag}(\Sigma_\ell), \quad C_\ell \equiv D_\ell^{-1/2} \Sigma_\ell D_\ell^{-1/2}, \quad (50)$$

one obtains the normalized-correlation shrinkage identity

$$\tilde{C}_\ell = \rho_\ell C_\ell + (1 - \rho_\ell) I, \quad (51)$$

which explicitly suppresses off-diagonal correlations and pushes the spectrum toward isotropy. A convenient quantitative proxy is the participation-ratio effective rank

$$R_{\text{eff}}(\Sigma) \equiv \frac{(\text{tr} \Sigma)^2}{\text{tr}(\Sigma^2)}. \quad (52)$$

Combining (49) with trace identities yields the closed form

$$R_{\text{eff}}(\tilde{\Sigma}_\ell) = \frac{(\text{tr} \Sigma_\ell)^2}{\rho_\ell^2 \text{tr}(\Sigma_\ell^2) + (1 - \rho_\ell^2) \text{tr}(\text{diag}(\Sigma_\ell)^2)}, \quad (53)$$

and since $\text{tr}(\Sigma_\ell^2) \geq \text{tr}(\text{diag}(\Sigma_\ell)^2)$ for any PSD Σ_ℓ , it follows that

$$R_{\text{eff}}(\tilde{\Sigma}_\ell) \geq R_{\text{eff}}(\Sigma_\ell), \quad (54)$$

with equality only when Σ_ℓ is diagonal. Thus dropout increases effective rank at the level of second moments by injecting entropy and suppressing correlation buildup.

This mechanism is inherently preventative: if the representation has already collapsed to an essentially one-dimensional signal subspace $\Sigma_\ell \approx vv^\top$, then (49) primarily adds diagonal variance rather than constructing new task-relevant modes, and under inverted dropout this injected variance is absent at inference. Consequently, at fixed dropout budget the mean-field propagation objective prefers step-like schedules but does not care about ordering, while rank flow breaks the permutation symmetry by favoring placing the saturated block where collapse is fastest. In regularization and rank-collapse limited regimes (especially in attention-dominated models), front-loading dropout acts as an entropic buffer that preserves higher-rank representations through the early layers, after which later layers can compress and refine a richer set of task-relevant features.

To test these predictions, we deliberately work in controlled regimes where mean-field assumptions remain valid (width

$\gg$ depth, moderate dropout) rather than pursuing benchmark performance. We validate on MLPs and Vision Transformers using CIFAR-10/100 (full details in App. A), where finite-width corrections and extensive data augmentation do not obscure the signal-propagation effects central to our analysis. Across architectures, early step-like schedules consistently outperform constant dropout (e.g., ViT on CIFAR-100: linearly decreasing achieves 49.38% vs 48.69% constant, $p < 0.05$), confirming that the theoretically-predicted scheduling principle transfers beyond the MLP setting in which it was derived.

4. Conclusions and further scope for research

Starting from mean-field theory, we showed that the analytic structure of activation functions defines universality classes for information propagation under dropout, with smooth and kinked nonlinearities exhibiting distinct scaling laws near criticality. The central contributions are the Landau equation of state for the decorrelation order parameter, the two-parameter scaling collapse, and the identification of distinct critical exponents across universality classes. As a demonstration that the framework has practical predictive power, we derived dropout schedules from the theory and validated them experimentally; the predicted front-loaded schedules consistently outperform constant dropout. Extending this mean-field analysis to attention mechanisms and incorporating finite-width corrections are natural directions for future work.

Impact statement

This paper’s contribution can benefit the general ML community through lower costs associated to training for equal performance, but has no further ethical implications.

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A. Experimental parameters and figures

In this section we explore some experimental results to support our theory. The code used to derive them, and all other figures in this paper are available in an attached ZIP file.

A.1. Mean-field theory and Scaling collapse

Here, we present the experimentally extracted exponents obtained by iteration of the MFT recursions 62.

Exponent	Iterated recursion	Linear approximation
ν	1.0190 ± 0.0014	1
$\beta_{\tanh}$	0.9879 ± 0.0007	1
β_{ReLU}	1.9642 ± 0.0019	2
$p_{\tanh}$	1.0048 ± 0.0000	1
p_{ReLU}	1.9870 ± 0.0001	2
$1/\delta_{\tanh}$	0.5171 ± 0.0007	1/2
$1/\delta_{\text{ReLU}}$	0.6656 ± 0.0001	2/3
$\nu_{\rho, \tanh}$	0.4716 ± 0.0014	1/2
$\nu_{\rho, \text{ReLU}}$	0.3528 ± 0.0010	1/3

Table 3. Critical exponents from log–log linear fits (dashed lines) in Fig. 1 compared with mean-field theory predictions.

Additionally, we show the scaling collapse counterpart for kinked activation functions, with ReLU as the choice of activation.

A.2. Dropout scheduling as an application

As a test of the mean-field framework’s predictive power, we examine whether the dropout schedules derived in Sec. 3.3 improve generalization in practice. Across architectures, regimes, and hyperparameter choices, we consistently found that front-loaded dropout, implemented via a step-like schedule concentrated in early layers, improves generalization and test accuracy relative to the schedules we tested: constant dropout, linearly increasing or decreasing dropout, step-like dropout concentrated in either the first or last half, and no dropout.

A simple interpretation is that the coadaptation that dropout is meant to counter (Wager et al., 2013) tends to accumulate early in the network, so injecting stochasticity early is often the most effective use of a fixed dropout budget. A more technical defense of this tie-breaking intuition comes from rank flow, and it is explored at length in the main body. There is a symmetrical scenario due to Eq. (37) being invariant under permutations of h_ℓ , so at the level of mean-field correlation dynamics the ordering of a schedule is a priori irrelevant; one must appeal to a beyond-mean-field symmetry-breaking mechanism. Empirically (Fig. 7), many schedules exhibit a test-loss minimum at intermediate dropout strength, consistent with a generalization bottleneck. We interpret this as a collapse into trivial representations, which we quantify via an effective-rank proxy. In MLPs this collapse is exponential (rather than doubly exponential as in pure-attention models (Dong et al., 2021)); specifically (Daneshmand et al., 2020), one expects the departure from one-dimensionality to decay at a rate set by the Lyapunov gap:

$$R(\ell) - 1 \sim (R(0) - 1) e^{(-2(\lambda_1 - \lambda_2)\ell)}, \quad (55)$$

where $\lambda_1 > \lambda_2$ are the top Lyapunov exponents governing singular-value growth. Additionally, we know dropout acts as a decorrelator by shrinking normalized correlations. Now we note that the second moment $\Sigma_\ell \equiv \mathbb{E}[h_\ell h_\ell^\top]$ transforms as

$$\tilde{\Sigma}_\ell = \Sigma_\ell + \left(\frac{1}{\rho_\ell} - 1\right) \text{diag}(\Sigma_\ell), \quad (56)$$

which inflates variances while leaving cross-covariances unchanged. Equivalently, the associated correlation matrix

$$C_\ell \equiv D_\ell^{-1/2} \Sigma_\ell D_\ell^{-1/2}, \quad D_\ell \equiv \text{diag}(\Sigma_\ell), \quad (57)$$

obeys the simple shrinkage rule

$$\tilde{C}_\ell = \rho_\ell C_\ell + (1 - \rho_\ell) I, \quad (58)$$

Schedule	ξ_{eff}	L/ξ_{eff}	Best Test Accuracy (%)
Constant	3.20	1.87	42.54 ± 0.10
Step (late)	5.09	1.18	40.56 ± 0.11
Step (early)	5.09	1.18	43.13 ± 0.07
Linear (increasing)	3.73	1.61	41.01 ± 0.07
Linear (decreasing)	3.73	1.61	43.11 ± 0.09
No dropout	∞	0	38.77 ± 0.10

Table 4. Performance of different dropout schedules in the overfitting regime for MLPs and CIFAR-10, corresponding to Fig. 5.

explicitly suppressing off-diagonal correlations and thereby tending to increase effective rank (by making the spectrum more uniform). Crucially, this mechanism is most useful before the representation becomes effectively one-dimensional in its signal spectrum: once a single mode dominates, injecting additional diagonal/noise variance does not recreate missing task-relevant directions and is averaged out at inference. Late-dropout schedules therefore allow early collapse to proceed largely unimpeded and then inject noise too late to preserve diversity. Conversely, early dropout acts as an entropic buffer, preserving higher-rank representations through the critical early layers and allowing compression to occur later, when richer task-relevant features can form.

Motivated by these findings, and the understanding that rank collapse occurs at an even faster rate in attention networks, we applied the same scheduling intuition to Vision Transformers (ViT), placing dropout in both attention and MLP sublayers within each transformer block. We compared no dropout, constant dropout, step-like schedules, and linearly decreasing schedules (heavier early). The findings in Fig. 7 and App. A give experimental confirmation of the theoretical predictions in a controlled regime, and open the door for further theoretical exploration of MFT in modern architectures.

A.3. MLPs

Here we show experiments validating the theoretical predictions on dropout scheduling. The main finding is that concentrating dropout in early layers, where coadaptation and rank collapse are most severe, yields the best generalization. We demonstrate that the theoretically-motivated scheduling improves the baseline constant scheduling at no extra computational cost.

To test this prediction in a controlled finite-width setting, we design an experiment in the overfitting regime, where information propagation and regularization are both important factors in determining performance. In the effective-theory treatment of (Roberts et al., 2022), deviations from Gaussian mean-field behavior (Neal, 1996; Lee et al., 2018) generate corrections to the correlation recursions that are suppressed by $1/N$ (with N the layer width) and accumulate across depth as L/N . At the same time, deep information propagation suggests that trainable depth extends only to a modest multiple of the correlation length; empirically, one often finds successful learning out to $\mathcal{O}(6)\xi$ (Schoenholz et al., 2017).

We train on 5000 CIFAR-10 images with width 256 and depth 2ξ , using a batch size of 75 to ensure the regularizing effect of dropout through injected stochastic noise. We initialize close to the edge-of-chaos at $\sigma_w^2 = 1.98$ and $\sigma_b^2 = 0.02$, near standard He initialization but avoiding the instabilities due to the divergence of the variance fixed point q^* as $\sigma_w^2 \rightarrow 2$.

We compare six depth profiles at fixed mean dropout budget: (i) no dropout, (ii) uniform dropout, (iii) a linearly increasing schedule, (iv) a linearly decreasing schedule, (v) a step schedule concentrated in the final half (step), and (vi) a step schedule concentrated in the first half (early step), as summarized in Fig. 5. Front-loaded dropout consistently outperforms back-loaded dropout: the need for regularization in early layers, where coadaptation builds up most rapidly, dominates the signal propagation cost. The effective correlation length heuristic remains predictive; constant dropout is most damaging, but rank collapse breaks the permutation symmetry of Eq. (39) in favor of early regularization.

A natural objection to the above results is that the improved performance of early-concentrated dropout might simply reflect the network’s need for stronger local regularization, rather than any benefit from spatial allocation. After all, the step schedule applies dropout at rate $2\bar{h}$ to the first half of the network, so perhaps the same improvement could be achieved by applying this higher rate uniformly throughout.

To disentangle these effects, we compare schedules where dropout is constant, with value $\bar{h}$, $2\bar{h}$, $3\bar{h}$, and where we have a smaller step of $2\bar{h}$ for the first half of the network, and one with a bigger step ($3\bar{h}$) for the first third of the network, with results in Fig. 6 and Tab. 5. If the step schedules succeed merely because they apply locally higher dropout rates, then the

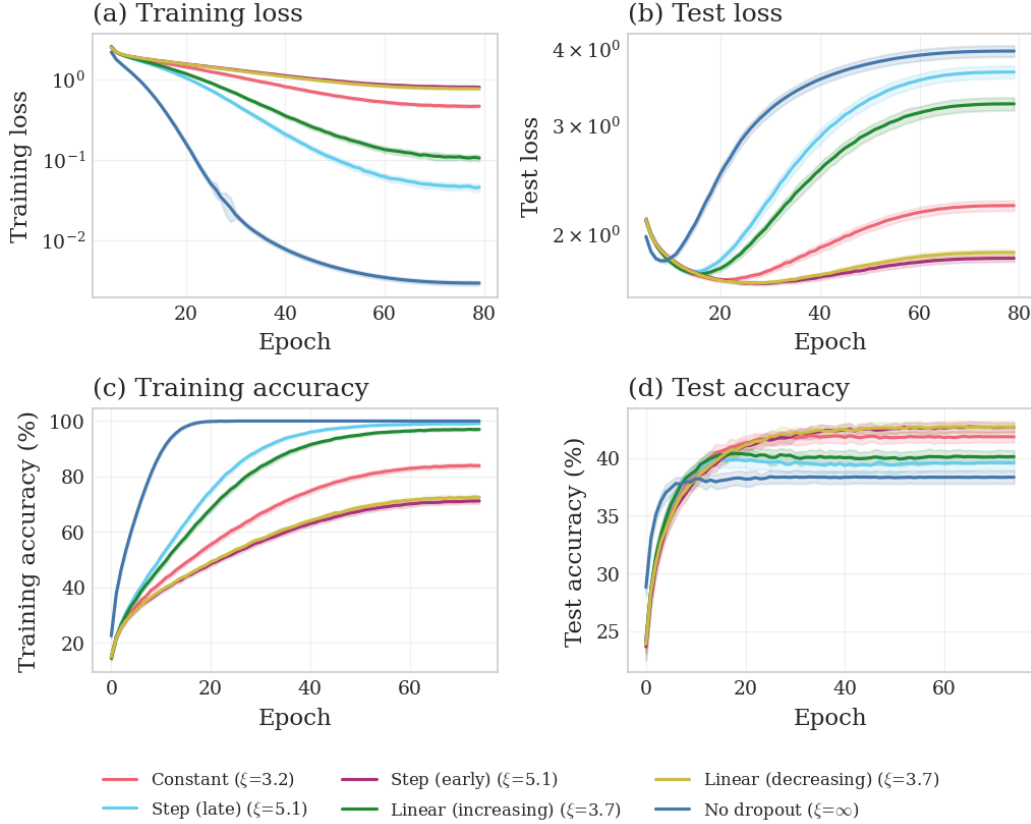


Figure 5. Depth-dependent dropout schedules used in the finite-width experiment at fixed mean dropout budget $\bar{h}$ (with $0 \leq h_\ell \leq h_{\max}$). We compare uniform dropout, linear ramps (increasing/decreasing with depth), and step schedules that concentrate dropout in either the first or last half of the network, together with the no-dropout baseline.

uniform schedules with matching dropout should perform at least as well. In contrast, if spatial allocation genuinely matters, the step schedules should outperform their uniform counterparts despite having lower total dropout. The results in Fig. 6 and Table 5 confirm that spatial allocation dominates: Big step ($\xi_{\text{eff}} = 6.7$) achieves the highest accuracy despite having only one-third the total dropout of the schedule with $3\bar{h}$, and front-loaded step ($\xi_{\text{eff}} = 5.1$) outperforms Double despite half the total dropout budget. Moreover, increasing uniform dropout beyond $2\bar{h}$ becomes counterproductive; Triple ($3\bar{h}$) performs worst among all dropout schedules, as excessive regularization throughout the network impairs learning capacity. These findings validate the effective correlation length ξ_{eff} as a predictive metric: schedules with longer ξ_{eff} consistently achieve better generalization, regardless of their total dropout budget.

We note that our theoretical analysis is perturbative in h , with higher-order corrections of $O(h^2)$ becoming significant as dropout increases. This motivates our choice to test up to $3\bar{h}$ with $\bar{h} = 0.1$; beyond this point, the perturbative framework becomes less reliable and additional nonlinear effects may dominate. We also tested these effects for lower dropout rates obtaining similar results, though they become more pronounced as dropout increases.

Schedule	ξ_{eff}	L/ξ_{eff}	Best Test Accuracy (%)
No dropout	∞	0.00	38.77 ± 0.10
Constant	3.20	1.87	42.54 ± 0.10
Step (early)	5.09	1.18	43.13 ± 0.07
Big step (1/3)	6.67	0.90	43.29 ± 0.08
Double ($2\bar{h}$)	2.54	2.36	41.46 ± 0.08
Triple ($3\bar{h}$)	2.22	2.70	33.94 ± 0.22

Table 5. Summary of dropout schedules, correlation lengths, and test performance for MLPs on CIFAR-10, corresponding to Fig. 6.

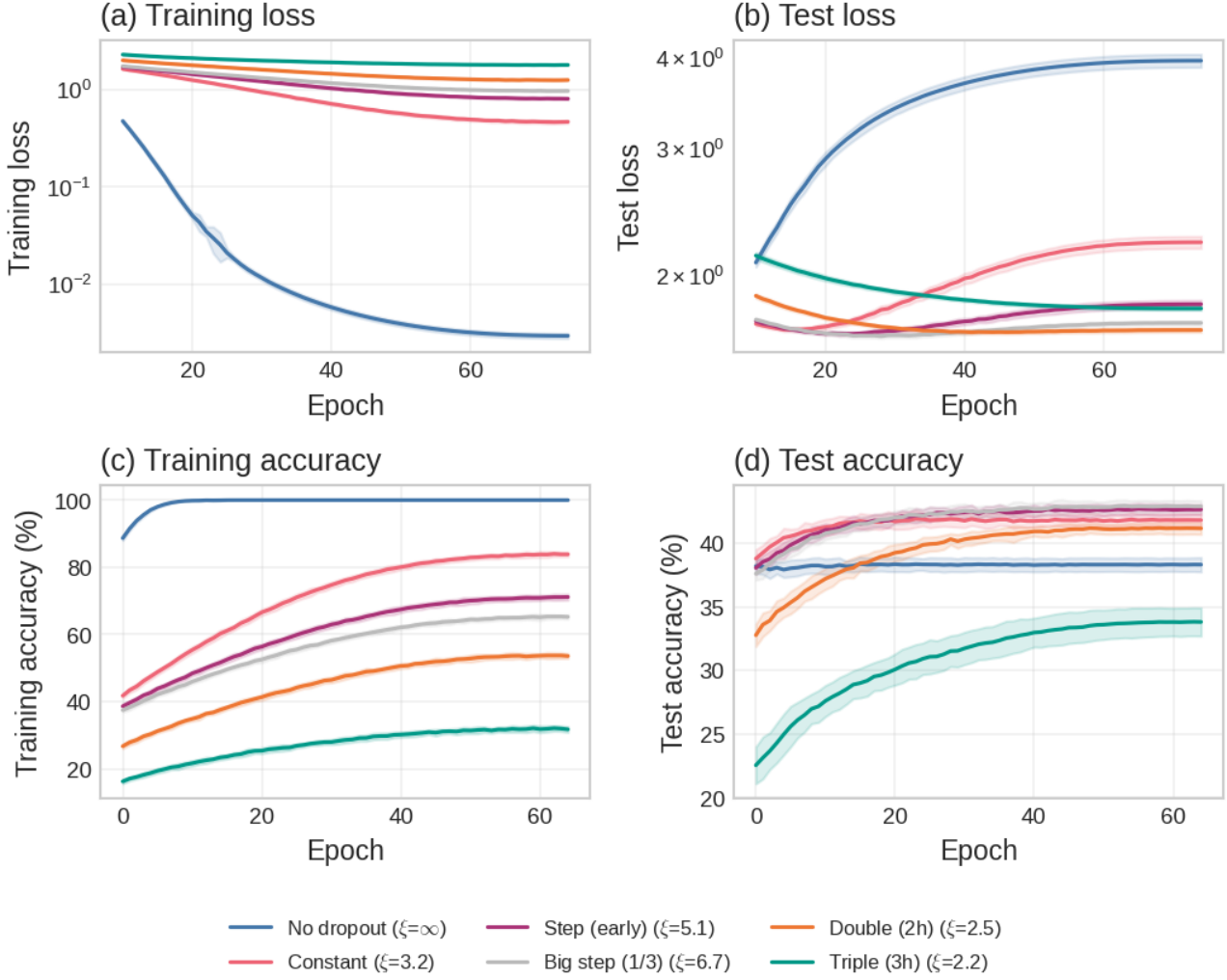


Figure 6. We test different dropout schedules. Three of them have the same average dropout throughout, with a step of twice and thrice the amount of dropout as we would find in the constant dropout case. We also test using constant dropout with twice and thrice the amount of dropout used in the base case.

A.4. Transformers CIFAR-100

Here we gather the hyperparameters and architecture used to perform the main transformer experiment, as shown in Fig. 7, with results in Tab. 8.

Parameter	Value / Description
<i>Network Architecture & Initialization</i>	
Network Width	256
Depth Multiplier	2
Weight Variance (σ_w^2)	1.98
Bias Variance (σ_b^2)	0.02
<i>Training Dynamics</i>	
Epochs	50
Batch Size	75
Learning Rate	1×10^{-4}
LR Schedule	Decay to minimum 1×10^{-7}
Mean Dropout Field ($\bar{h}$)	0.1
<i>General Setup</i>	
Number of Simulations	20
Dataset Usage	CIFAR-10 5,000 samples
Hardware	GPU A100

Table 6. Experimental and hyperparameter configuration for high overfitting regime. Corresponding performance metrics in Fig. 5.

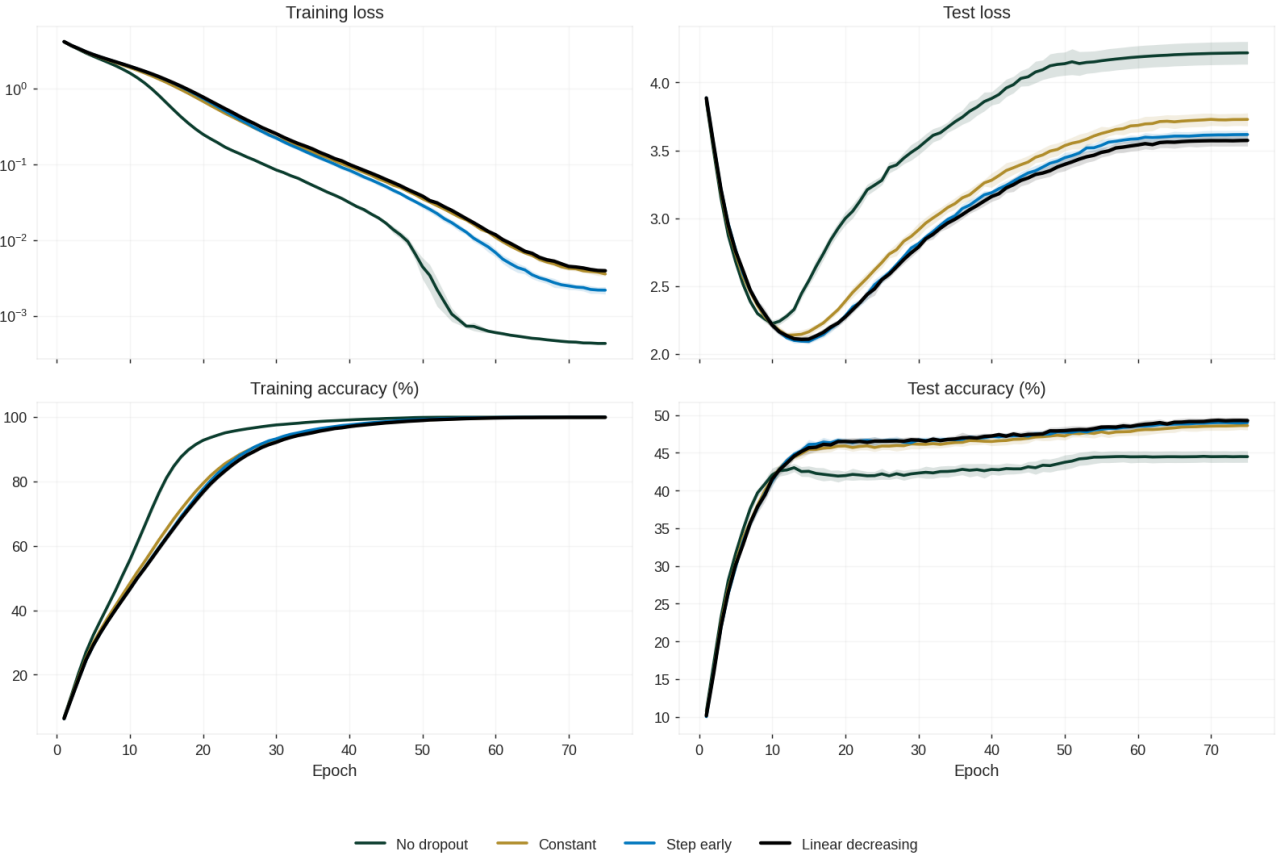


Figure 8. Extended training curves for Figure 7, showing the full 75 epochs.

Accuracy curves are shown in Fig. 7, with explicit results in Table 8. We emphasize that all errors in the tables displayed

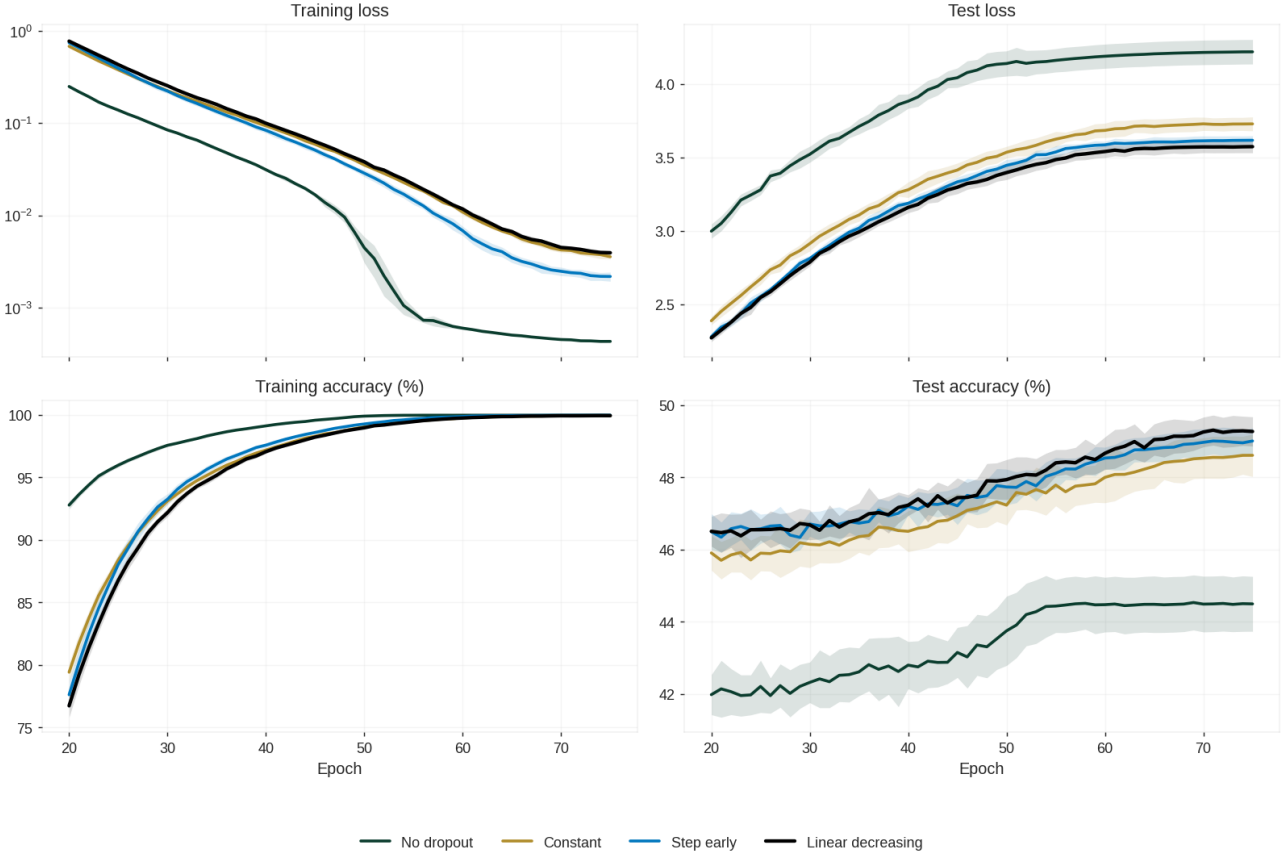


Figure 7. Depth-dependent dropout schedules used in the finite-width experiment at fixed mean dropout budget $\bar{h}$. We compare no uniform, linear ramps (decreasing with depth), and step dropout schedules, together with the no-dropout baseline. We show epochs 20-75 to make the results clearer, the uncropped version is also in App. A.

in this paper are SEM. On CIFAR-100 (and similarly on CIFAR-10), concentrating dropout earlier consistently improved accuracy and reduced overfitting. Interestingly, reverse linear (heavier early) schedules often performed especially well in transformers; we suspect residual connections (He et al., 2016) ease information propagation enough that distributing regularization early can dominate the propagation cost, though both early weighted and step-like schedules outperformed the constant baseline.

We focus on controlled regimes rather than ImageNet-scale benchmarking because our central claims are mechanistic and derived in a mean-field initialization framework, which is most cleanly tested when finite-width corrections remain subdominant. The experimental validation is designed to test the theory in settings where MFT predictions apply cleanly, rather than to establish state-of-the-art performance. ImageNet (Deng et al., 2009) training introduces substantial confounders from augmentation and extensive hyperparameter tuning that can obscure MFT effects; we leave ImageNet evaluation as future work. Overall, the predicted schedules consistently outperform constant dropout, illustrating the practical utility of the mean-field framework.

A.5. Transformer ablations

To isolate the effect of dropout scheduling on distinct transformer components, we performed component-level ablations. We tested transformers with dropout applied exclusively to the attention block, the MLP block, or both, comparing step-like and constant schedules against a no-dropout baseline. Experimental hyperparameters are listed in Table 9; they largely mirror the main transformer experiment, though we conducted five simulations per configuration. Additionally, we validated these theoretical results on a second dataset, CIFAR-10.

Parameter	Value / Description
<i>ViT Architecture</i>	
Embedding Dimension	128
Depth	12
Number of Heads	16
MLP Ratio	4.0
Patch Size	4
<i>Training Dynamics</i>	
Epochs	75
Batch Size	250
Learning Rate	5×10^{-4}
LR Minimum	5×10^{-6}
Weight Decay	0.00
Mean Dropout Field ($\bar{h}$)	0.1
Max Dropout Field (h_{max})	0.2
Schedules	None, Constant, Reverse Step, Reverse Linear
<i>General Setup</i>	
Number of Simulations	10
Dataset Usage	Full CIFAR-100
Evaluation Frequency	Every 1 Epoch
Hardware	GPU A100

Table 7. Experimental configuration for Vision Transformer (ViT) simulations.

Schedule	Best Test Acc	Final Test Acc
No dropout	44.61 ± 0.25	44.50 ± 0.25
Constant	48.69 ± 0.18	48.61 ± 0.20
Step (early)	49.10 ± 0.11	49.01 ± 0.12
Linear (decreasing)	49.38 ± 0.14	49.27 ± 0.14

Table 8. Test accuracy across different dropout schedules for ViT architecture on CIFAR-100, corresponding to Fig. 7.

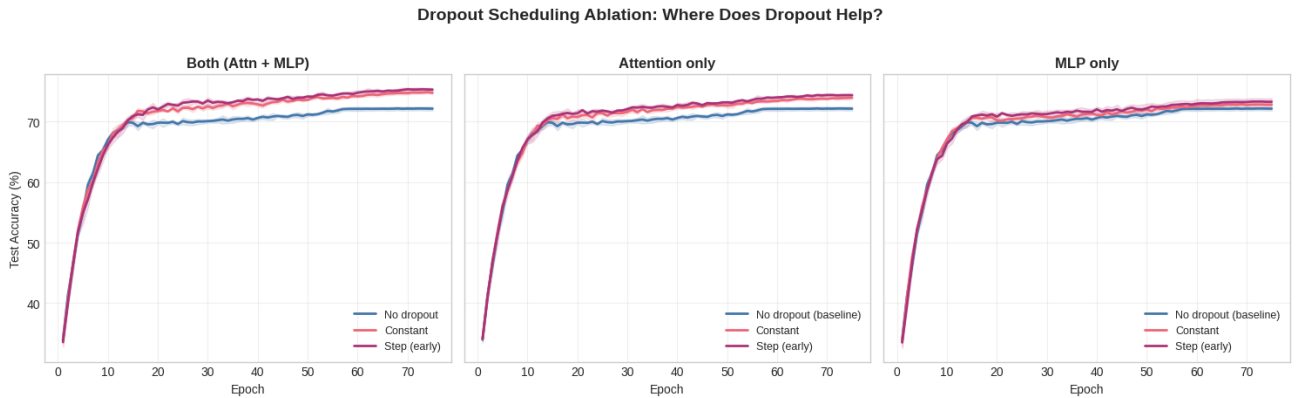


Figure 9. We present the performance of our models under different ablations, where the dropout scheduling is applied to either the Attention layers, the MLP layers, or both. The relative performance of no dropout, constant, and step-like dropout is shown.

Parameter	Value / Description
<i>ViT Architecture</i>	
Embedding Dimension	128
Depth	12
Number of Heads	16
MLP Ratio	4.0
Patch Size	4
<i>Training Dynamics</i>	
Epochs	75
Batch Size	1000
Learning Rate	5×10^{-4}
LR Minimum	5×10^{-6}
Weight Decay	0.00
Mean Dropout Field ($\bar{h}$)	0.1
Max Dropout Field ($h_{\max}$)	0.2
Schedules	None, Constant, Reverse Step
Ablation Modes	Both, Attention-only, MLP-only
<i>General Setup</i>	
Number of Simulations	5
Dataset Usage	Full CIFAR-10
Evaluation Frequency	Every 1 Epoch
Hardware	GPU A100

Table 9. Experimental configuration for Vision Transformer (ViT) ablation study.

Configuration	Best Test Acc	Final Test Acc
No dropout	72.22 ± 0.31	72.15 ± 0.33
MLP only (Constant)	72.96 ± 0.26	72.88 ± 0.28
MLP only (Step early)	73.38 ± 0.56	73.29 ± 0.55
Attn only (Constant)	74.00 ± 0.20	73.95 ± 0.21
Attn only (Step early)	74.48 ± 0.20	74.41 ± 0.19
Both (Constant)	74.92 ± 0.22	74.78 ± 0.23
Both (Step early)	75.47 ± 0.27	75.39 ± 0.29

Table 10. Component-level ablations on CIFAR-10, comparing Constant and Step-like dropout schedules applied exclusively to the MLP block, Attention block, or both. The Step (early) schedule consistently outperforms the Constant schedule across all configurations.

B. Mean-field theory primer

Here we briefly review the mean-field formalism of deep information propagation (DIP) (Schoenholz et al., 2017), which will serve as the baseline for what follows. The context is a fully-connected MLP at random initialization, studied in the wide limit, where preactivations are well-approximated as Gaussian random variables with self-consistently determined covariances. In the strict $N_l \rightarrow \infty$ limit this Gaussian description closes, while at large but finite width deviations from Gaussianity can be organized perturbatively in the depth-to-width aspect ratio (schematically L/N), as emphasized in (Roberts et al., 2022).

We consider a depth- L network with N_l Neurons in layer l , and an activation $\phi : \mathbb{R} \rightarrow \mathbb{R}$ (e.g. tanh, ReLU, etc.). We use

the standard i.i.d. Gaussian initialization

$$W_{ij}^l \sim \mathcal{N}\left(0, \frac{\sigma_w^2}{N_l}\right), \quad b_i^l \sim \mathcal{N}(0, \sigma_b^2), \quad (59)$$

take the input as $y_{i;a}^0 = x_{i;a}$, and propagate forward via

$$z_{i;a}^l = W_{ij}^l y_{j;a}^l + b_i^l, \quad y_{i;a}^{l+1} = \phi(z_{i;a}^l), \quad (60)$$

where a labels the input. Throughout, expectations $\mathbb{E}[\cdot]$ are over the random weights and biases at initialization, with inputs held fixed. We also use the standard Gaussian measure

$$\int Dz(\cdots) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz e^{-z^2/2}(\cdots),$$

and similarly for $\int Dz_1 Dz_2$.

The mean-field approximation is that, at large width, each preactivation is a sum of many weakly-dependent terms and is therefore well modelled by a Gaussian with matching first two moments. In the infinite-width limit this becomes exact layer-by-layer, while at finite width the first layer is exactly Gaussian (being a linear map of fixed inputs) and subsequent layers are only approximately Gaussian, with controlled non-Gaussian corrections.

We first track the layer-wise preactivation variance for a single input a ,

$$q_{aa}^l := \mathbb{E}\left[(z_{i;a}^l)^2\right], \quad \mathbb{E}[z_{i;a}^l] = 0. \quad (61)$$

At initialization, q_{aa}^l is deterministic in MFT and evolves by a closed recursion. Conditioning on the previous layer and using independence and zero means gives

$$q_{aa}^l = \sigma_w^2 \int Dz \phi^2\left(\sqrt{q_{aa}^{l-1}} z\right) + \sigma_b^2. \quad (62)$$

When it exists, the large-depth fixed point is

$$q^* \equiv \lim_{l \rightarrow \infty} q_{aa}^l. \quad (63)$$

For bounded activations (e.g. tanh) the fixed point exists for any σ_w^2, σ_b^2 . For unbounded activations (e.g. ReLU) it exists only in part of parameter space; outside that region the recursion can run away and the typical preactivation scale diverges with depth.

We now consider the correlations between two distinct inputs a, b . Mean-field predicts that different Neurons decorrelate at leading order, so we define

$$\mathbb{E}[z_{i;a}^l z_{j;b}^l] = q_{ab}^l \delta_{ij}. \quad (64)$$

The corresponding mean-field recursion is

$$q_{ab}^l = \sigma_w^2 \int Dz_1 Dz_2 \phi(u_1) \phi(u_2) + \sigma_b^2, \quad (65)$$

with

$$u_1 = \sqrt{q_{aa}^{l-1}} z_1, \quad c_{ab}^{l-1} = \frac{q_{ab}^{l-1}}{\sqrt{q_{aa}^{l-1} q_{bb}^{l-1}}}, \quad (66)$$

$$u_2 = \sqrt{q_{bb}^{l-1}} \left(c_{ab}^{l-1} z_1 + \sqrt{1 - (c_{ab}^{l-1})^2} z_2 \right). \quad (67)$$

After the variance dynamics has relaxed to a common fixed point q^* (so $q_{aa}^l = q_{bb}^l = q^*$), this induces a one-dimensional correlation map $c_{ab}^l = F(c_{ab}^{l-1})$. In the absence of dropout, perfect alignment is always a fixed point: $c^* = 1$ solves $c^* = F(c^*)$.

To probe its stability, one linearizes the map at $c = 1$, defining the DIP susceptibility

$$\chi_1 \equiv \left. \frac{\partial c_{ab}^l}{\partial c_{ab}^{l-1}} \right|_{c_{ab}=1} = \sigma_w^2 \int Dz [\phi'(\sqrt{q^*} z)]^2. \quad (68)$$

(In the language of (Roberts et al., 2022), this is the angular susceptibility $\chi_{\perp}$.) This creates a phase diagram with three regimes:

- $\chi_1 < 1$: ordered regime. The fixed point $c^* = 1$ is attractive: different inputs become asymptotically aligned, and gradients typically vanish with depth.
- $\chi_1 > 1$: chaotic regime. Perfect alignment is unstable, correlations flow to a nontrivial fixed point $c^* < 1$, and gradients typically explode with depth.
- $\chi_1 = 1$: critical regime (the edge-of-chaos). The depth scale controlling cross-input correlations can become arbitrarily large.

To show the power of criticality, we will derive a well known initialization based purely on demanding criticality: He (He et al., 2015). Let us take $\phi(z) = \max(0, z)$, so that $\phi'(z) = \theta(z)$ and

$$\int Dz (\phi'(\sqrt{q^*} z))^2 = \int Dz \theta(z) = \frac{1}{2}. \quad (69)$$

Hence

$$\chi_1 = \frac{\sigma_w^2}{2}, \quad \chi_1 = 1 \implies \sigma_w^2 = 2, \quad (70)$$

which is precisely the He variance condition (variance $2/N$) for ReLU. At the same time, the ReLU variance recursion simplifies to

$$q_{aa}^l = \frac{\sigma_w^2}{2} q_{aa}^{l-1} + \sigma_b^2,$$

so for $\sigma_b^2 > 0$ the fixed point is $q^* = \sigma_b^2 / (1 - \sigma_w^2/2)$ when $\sigma_w^2 < 2$ and ceases to exist at $\sigma_w^2 \geq 2$ (diverging as $\sigma_w^2 \rightarrow 2^-$). Thus, if one insists on $\sigma_b^2 > 0$, tuning exactly to $\chi_1 = 1$ is incompatible with a finite q^* for ReLU, and in practice one often works slightly subcritical (or sets biases to zero) to keep the signal scale under control. For smooth bounded ϕ , changing σ_b^2 shifts q^* and therefore shifts the typical slope $\phi'(\sqrt{q^*} z)$, so the critical value of σ_w^2 becomes σ_b^2 -dependent.

Beyond the phase diagram, these recursions "grow" characteristic depth scales. Iterating random affine maps induces an effective coarse-graining in depth akin to the renormalization group flow in theoretical physics, coined representation group flow in the ML context (Roberts et al., 2022): information about norms and angles relaxes exponentially fast toward their fixed points. Concretely, perturb the variance around q^* by an infinitesimal amount,

$$q_{aa}^l = q^* + \epsilon_q^l. \quad (71)$$

Expanding (62) to first order yields

$$\epsilon_q^{l+1} = \epsilon_q^l \left[\chi_1 + \sigma_w^2 \int Dz \phi''(\sqrt{q^*} z) \phi(\sqrt{q^*} z) \right]. \quad (72)$$

Defining

$$r_q \equiv \chi_1 + \sigma_w^2 \int Dz \phi''(\sqrt{q^*} z) \phi(\sqrt{q^*} z), \quad (73)$$

we have asymptotically

$$\epsilon_q^l = \epsilon_q^0 r_q^l = \epsilon_q^0 e^{-l/\xi_q}, \quad \xi_q^{-1} \equiv -\log |r_q|. \quad (74)$$

Similarly, perturbing the correlation around its fixed point c^* gives an exponential approach with depth scale

$$\xi_c^{-1} = -\log \left[\sigma_w^2 \int Dz_1 Dz_2 \phi'(u_1^*) \phi'(u_2^*) \right], \quad (75)$$

$$u_1^* = \sqrt{q^*} z_1, \quad u_2^* = \sqrt{q^*} (c^* z_1 + \sqrt{1 - (c^*)^2} z_2). \quad (76)$$

Heuristically, ξ_c controls how far distinctions between different inputs survive with depth. In the ordered phase $c^* = 1$, so $u_1^* = u_2^*$ and $\xi_c^{-1} = -\log \chi_1$, hence $\chi_1 \rightarrow 1$ implies $\xi_c \rightarrow \infty$. Meanwhile, ξ_q controls how quickly single-input norms settle to q^* . For smooth saturating activations such as $\tanh$, the curvature term is typically negative, so even at $\chi_1 = 1$ one often finds $|r_q| < 1$ and ξ_q stays finite and relatively short.

In this sense, the edge-of-chaos is primarily about ξ_c rather than ξ_q , except in special cases. For ReLU, $\phi'' = 0$ away from the origin (so the curvature contribution vanishes in the Gaussian averages), and one has $r_q = \chi_1$. In that case ξ_q and ξ_c essentially coincide and both diverge as $\chi_1 \rightarrow 1$, provided a finite q^* exists. This is one concrete reason why ReLU-style near-critical initializations are comparatively forgiving in practice.

MFT provides a good approximation in very wide networks, especially through the first few layers. At large but finite width, deviations from Gaussian mean-field behavior can be organized perturbatively in the depth-to-width aspect ratio L/N . Accordingly, in our experiments we work in a controlled regime with $L/N \ll 1$ (i.e., $N \gg L$). Near criticality, the correlation length becomes large, which slows relaxation to the fixed point and typically amplifies sensitivity to finite-width and other non-Gaussian corrections. Hence, when probing critical power laws while maintaining dropout, we keep the dropout rate parametrically small but non-negligible: taking dropout to be infinitesimal would require prohibitively deep networks to obtain precise estimates of the correlation length and related quantities, making the experiments computationally expensive.

B.1. Derivation of correlation map and derivatives with dropout

This subsection supplies the steps leading to Eqs. (15) and (16), starting from the two-input correlation recursion that underlies the perfect-alignment shift $\bar{F}_\rho(1) = 1 - \Delta$ in (13).

Fix a layer where the single-input variance has converged to the dropout-shifted fixed point $\bar{q}^*$ defined by (11). For two inputs, let (u_1, u_2) be jointly Gaussian with

$$\mathbb{E}[u_1^2] = \mathbb{E}[u_2^2] = \bar{q}^*, \quad \mathbb{E}[u_1 u_2] = c \bar{q}^*, \quad c \in [-1, 1].$$

The normalized correlation map is

$$\bar{F}_\rho(c) \equiv \frac{\bar{q}_{ab}^{l+1}}{\bar{q}^*} = \frac{\sigma_w^2 \mathbb{E}[\phi(u_1)\phi(u_2)] + \sigma_b^2}{\bar{q}^*}. \quad (77)$$

At $c = 1$, $u_1 = u_2$ and $\bar{F}_\rho(1) = 1 - \Delta$ by definition, giving (13).

Since $\bar{q}^*$ is independent of c , differentiating (77) yields

$$\bar{F}'_\rho(c) = \frac{\sigma_w^2}{\bar{q}^*} \frac{\partial}{\partial c} \mathbb{E}[\phi(u_1)\phi(u_2)]. \quad (78)$$

Write $u_i = \sqrt{\bar{q}^*} v_i$, where (v_1, v_2) are standard Gaussians with $\mathbb{E}[v_1 v_2] = c$, and define

$$G(c) \equiv \mathbb{E}[\phi(\sqrt{\bar{q}^*} v_1) \phi(\sqrt{\bar{q}^*} v_2)].$$

Price's theorem gives

$$\frac{d}{dc} G(c) = \mathbb{E} \left[\frac{\partial^2}{\partial v_1 \partial v_2} \left(\phi(\sqrt{\bar{q}^*} v_1) \phi(\sqrt{\bar{q}^*} v_2) \right) \right]. \quad (79)$$

A direct differentiation yields

$$\frac{\partial^2}{\partial v_1 \partial v_2} \left(\phi(\sqrt{\bar{q}^*} v_1) \phi(\sqrt{\bar{q}^*} v_2) \right) = \bar{q}^* \phi'(\sqrt{\bar{q}^*} v_1) \phi'(\sqrt{\bar{q}^*} v_2).$$

Therefore,

$$G'(c) = \bar{q}^* \mathbb{E}[\phi'(\sqrt{\bar{q}^*} v_1) \phi'(\sqrt{\bar{q}^*} v_2)].$$

Taking $c \rightarrow 1$ collapses $v_1 = v_2 = z$ with $z \sim \mathcal{N}(0, 1)$, hence

$$G'(1) = \bar{q}^* \int Dz [\phi'(\sqrt{\bar{q}^*} z)]^2.$$

Substituting into (78) gives

$$\bar{F}'_\rho(1) = \sigma_w^2 \int Dz [\phi'(\sqrt{q^*} z)]^2,$$

which is Eq. (15). Differentiating (78) once more:

$$\bar{F}''_\rho(c) = \frac{\sigma_w^2}{\bar{q}^*} \frac{\partial^2}{\partial c^2} \mathbb{E}[\phi(u_1)\phi(u_2)]. \quad (80)$$

Applying Price's theorem again to $f(v_1, v_2) = \phi'(\sqrt{q^*} v_1)\phi'(\sqrt{q^*} v_2)$ yields

$$\frac{d}{dc} \mathbb{E}[\phi'(\sqrt{q^*} v_1)\phi'(\sqrt{q^*} v_2)] = \bar{q}^* \mathbb{E}[\phi''(\sqrt{q^*} v_1)\phi''(\sqrt{q^*} v_2)].$$

Combining with the previous step gives

$$\frac{\partial^2}{\partial c^2} \mathbb{E}[\phi(u_1)\phi(u_2)] = (\bar{q}^*)^2 \mathbb{E}[\phi''(\sqrt{q^*} v_1)\phi''(\sqrt{q^*} v_2)].$$

Evaluating at $c = 1$ again collapses $v_1 = v_2 = z$, so

$$\left. \frac{\partial^2}{\partial c^2} \mathbb{E}[\phi(u_1)\phi(u_2)] \right|_{c=1} = (\bar{q}^*)^2 \int Dz [\phi''(\sqrt{q^*} z)]^2.$$

Substituting into (80) yields

$$\bar{F}''_\rho(1) = \sigma_w^2 \bar{q}^* \int Dz [\phi''(\sqrt{q^*} z)]^2,$$

which is Eq. (16).

C. Derivation of critical exponents for smooth and kinked activation functions

C.1. Smooth Networks

Here we show that the standard static exponents follow directly from (19) with some extra steps and care taken. Along the critical isotherm $t = 0$,

$$h = \frac{g_\rho}{2} m^2 \quad \Rightarrow \quad m \sim h^{1/2},$$

so $\delta = 2$. The analogue of the magnetic susceptibility is $\chi_M(t) \equiv \partial m / \partial h|_{h \rightarrow 0}$. On the subcritical side $t < 0$, the analytic branch satisfies $m \simeq -h/t$, hence $\chi_M(t) \sim |t|^{-1}$ and $\gamma = 1$. Along $h = 0$, the physical branch is $m = 0$ for $t \leq 0$ and $m = 2t/g_\rho$ for $t \geq 0$, so $m \sim t$ as $t \rightarrow 0^+$ and $\beta = 1$.

The depth correlation length follows from linearizing the correlation dynamics about the displaced fixed point. To leading order,

$$\lambda \equiv \bar{F}'_\rho(c^*) = \bar{F}'_\rho(1 - m) \simeq \chi_\rho - g_\rho m = 1 + t - g_\rho m,$$

and substituting (20) gives

$$\lambda(t, h) = 1 - \sqrt{t^2 + 2g_\rho h}. \quad (81)$$

For $\lambda \simeq 1$ with $\lambda > 0$, the depth correlation length satisfies

$$\xi_{c,\rho}^{-1} \equiv -\log \lambda \simeq 1 - \lambda = \sqrt{t^2 + 2g_\rho h}.$$

Thus $\xi_{c,\rho}(t, 0) \sim |t|^{-1}$ and $\nu_t = 1$, while $\xi_{c,\rho}(0, h) \sim h^{-1/2}$ and $\nu_\rho = 1/2$. Equivalently, $y_t = 1/\nu_t = 1$ and $y_\rho = 1/\nu_\rho = 2$, so dropout is the more relevant perturbation of the correlation edge.

A pseudo-free-energy functional consistent with (19) is

$$f(t, m; h) = -\frac{t}{2} m^2 + \frac{g_\rho}{6} m^3 - h m, \quad (82)$$

since $\partial f/\partial m = 0$ is equivalent to (19). At $h = 0$, substituting the physical branch $m = 0$ for $t \leq 0$ and $m = 2t/g_\rho$ for $t \geq 0$ yields

$$f_{\text{on}}(t, 0) = 0 \quad (t \leq 0), \quad f_{\text{on}}(t, 0) = -\frac{2}{3g_\rho^2} t^3 \quad (t \geq 0).$$

Therefore the analogue of the specific heat, $C \equiv -\partial_t^2 f_{\text{on}}$, stays finite and vanishes linearly:

$$C(t) = \frac{4}{g_\rho^2} t \Theta(t) \sim |t|,$$

where $\Theta(t)$ is the Heaviside step function. In standard exponent language $C_{\text{sing}}(t) \sim |t|^{-\alpha}$, this corresponds to

$$\alpha = -1, \tag{83}$$

up to an additive analytic background $f_0(t, h)$ that does not affect the equation of state.

Collecting exponents for the smooth universality class:

$$\beta = 1, \quad \gamma = 1, \quad \delta = 2, \quad \nu_t = 1, \quad \nu_\rho = \frac{1}{2}, \quad \alpha = -1.$$

For comparison, standard Ising-like Landau mean-field theory (with $m \rightarrow -m$ symmetry and a quartic interaction) has

$$\beta = \frac{1}{2}, \quad \gamma = 1, \quad \delta = 3, \quad \nu = \frac{1}{2}, \quad \alpha = 0.$$

Our dropout-deformed correlation-edge theory shares $\gamma = 1$, but the lack of a $\mathbb{Z}_2$ symmetry (and the resulting cubic interaction) shifts β and δ , produces $\nu_t = 1$ rather than $\nu = 1/2$, and yields a specific-heat analogue that vanishes at the transition ($\alpha = -1$) rather than exhibiting the Ising mean-field behavior ($\alpha = 0$).

C.2. Networks with kinks

Up to now, the Landau-style expansion of $\bar{F}_\rho(c)$ around $c = 1$ relied on the assumption that $\bar{F}_\rho(c)$ is analytic in

$$m \equiv 1 - c.$$

For smooth activations this is the natural situation, and the first nonlinear correction is of order m^2 , controlled by

$$g_\rho \propto \int Dz [\phi''(\sqrt{q^*} z)]^2.$$

For kinked activations, this logic fails: the map remains smooth for $c < 1$, but as it approaches $c \rightarrow 1$, it is governed by a branch point, and the leading nonlinear correction is non-analytic in m . This changes the scaling of the dropout deformation.

We illustrate this explicitly for the ReLU activation $\phi(x) = \max(0, x)$. To keep formulas compact we set $\sigma_b = 0$ throughout this subsection and use the same inverted-dropout convention as before, with masks drawn independently across inputs.

The mean-field correlation map for ReLU admits a closed form. Writing $c \in [-1, 1]$ for the preactivation correlation at some depth,

$$F_{\text{ReLU}}(c) = \frac{1}{\pi} \left[\sqrt{1 - c^2} + (\pi - \cos^{-1} c) c \right]. \tag{84}$$

Near perfect alignment, setting $c = 1 - m$ with $0 < m \ll 1$, one obtains the non-analytic expansion

$$F_{\text{ReLU}}(1 - m) = 1 - m + \kappa m^{3/2} + \mathcal{O}(m^2), \tag{85}$$

$$\kappa \equiv \frac{2\sqrt{2}}{3\pi}.$$

And thus we see that the Taylor expansion for a smooth function would have missed the $m^{3/2}$ contribution.

We now turn on dropout and again view dropout as a field deformation at $c = 1$:

$$\bar{F}_\rho(1) = 1 - h, \quad h > 0.$$

In the present $\sigma_b = 0$ setting, this field is exactly

$$h = 1 - \bar{F}_\rho(1) = 1 - \rho, \quad (86)$$

independent of the activation, since at $c = 1$ the only source of mismatch is the independence of the two dropout masks. We write the fixed point as $c^* = 1 - m^*$ with $m^* > 0$ and impose $c^* = \bar{F}_\rho(c^*)$.

Near $m^* \ll 1$, the kinked analogue of the Landau equation of state is obtained by combining the shift $\bar{F}_\rho(1) = 1 - h$, the linear slope $\chi_\rho = 1 + t$, and the non-analytic term (85):

$$h = \kappa m^{3/2} - t m + \mathcal{O}(m^2), \quad t \equiv \chi_\rho - 1. \quad (87)$$

Along the critical isotherm $t = 0$, this gives

$$m^* \sim \left(\frac{h}{\kappa}\right)^{2/3}, \quad (t = 0), \quad (88)$$

so the smooth exponent $\delta = 2$ is replaced by

$$m \propto h^{1/\delta_{\text{kink}}}, \quad \delta_{\text{kink}} = \frac{3}{2}. \quad (89)$$

And with this we immediately find the smoking gun telling us that kinked and smooth activations fall into distinct mean-field universality classes under the dropout deformation. As before, the depth scale follows from linearizing the correlation dynamics around c^* . Differentiating (85) gives

$$\lambda \equiv \bar{F}'_\rho(c^*) \simeq 1 + t - \frac{3\kappa}{2} \sqrt{m^*}. \quad (90)$$

At $t = 0$ we therefore have

$$1 - \lambda \propto \sqrt{m^*} \propto h^{1/3}.$$

Using $\xi_{c,\rho}^{-1} \simeq 1 - \lambda$ for $\lambda \simeq 1$ yields

$$\xi_{c,\rho}(0, h) \sim h^{-1/3}, \quad \nu_{\rho, \text{kink}} = \frac{1}{3}, \quad (91)$$

to be compared with the smooth result $\xi_{c,\rho}(0, h) \sim h^{-1/2}$.

It is also useful, in direct analogy with the smooth case, to package (87) into a pseudo-free-energy. A functional whose stationarity condition reproduces the kinked equation of state is

$$f_{\text{kink}}(t, m; h) = -\frac{t}{2} m^2 + \frac{2\kappa}{5} m^{5/2} - h m, \quad (92)$$

since $\partial f_{\text{kink}}/\partial m = 0$ gives $h = \kappa m^{3/2} - t m$ at leading order.³ At zero field the physical branch is $m = 0$ for $t \leq 0$ and $m = (t/\kappa)^2$ for $t \geq 0$, so the on-shell free energy behaves as

$$f_{\text{kink, on}}(t, 0) = 0 \quad (t \leq 0), \quad f_{\text{kink, on}}(t, 0) = -\frac{1}{10\kappa^4} t^5 \quad (t \geq 0).$$

Thus the analogue of the specific heat, $C \equiv -\partial_t^2 f_{\text{kink, on}}$, stays finite and in fact vanishes as

$$C(t) = \frac{2}{\kappa^4} t^3 \Theta(t) \sim |t|^3.$$

³As before, f_{kink} is defined only up to addition of an m -independent analytic background $f_0(t, h)$, which does not affect the equation of state.

In the standard exponent convention $C_{\text{sing}}(t) \sim |t|^{-\alpha}$, this corresponds to

$$\alpha_{\text{kink}} = -3. \quad (93)$$

Although we used ReLU to compute the near-alignment expansion in closed form, the resulting fractional power is not a peculiarity of ReLU. Rather, it is a generic feature of the mean-field Gaussian channel whenever the activation has an ordinary kink: ϕ is continuous and piecewise C^1 with a finite jump in slope at some point. In that case the correlated-Gaussian expectation defining the normalized correlation map remains smooth for $c < 1$, but its approach to $c \rightarrow 1$ is controlled by a square-root branch associated with the thin Gaussian tube around the diagonal $u_1 = u_2$. Writing $c = 1 - m$ with $0 < m \ll 1$, the transverse fluctuations scale as $\sqrt{m}$, so the probability that two nearly aligned preactivations fall on opposite sides of the kink scales as $\mathcal{O}(\sqrt{m})$; in that straddling region a finite-slope kink produces activation mismatches of size $\mathcal{O}(\sqrt{m})$, so the leading non-linear correction to the kernel scales as $\mathcal{O}(\sqrt{m}) \times \mathcal{O}(m) = \mathcal{O}(m^{3/2})$. Thus, for any standard kinked activation one expects

$$F(1 - m) = 1 - m + \kappa_\phi m^{3/2} + \mathcal{O}(m^2), \quad (94)$$

with an activation-dependent coefficient κ_ϕ , while the exponent $3/2$ is fixed by the tube geometry.

Finally, it is useful to recall what happens when dropout is switched off exactly at the marginal point. Setting

$$\rho = 1, \quad t = 0,$$

the non-analytic correction controls the approach to $c = 1$. Writing $m_\ell \equiv 1 - c_\ell$ and using (85),

$$m_{\ell+1} - m_\ell \simeq -\kappa m_\ell^{3/2}. \quad (95)$$

In a continuum-depth approximation $m_{\ell+1} - m_\ell \rightarrow \partial_\ell m$, we obtain

$$\partial_\ell m \simeq -\kappa m^{3/2} \quad \Rightarrow \quad m(\ell) \sim \ell^{-2}, \quad (96)$$

which reproduces the characteristic $1 - c_\ell \sim \ell^{-2}$ relaxation of ReLU activations at the edge-of-chaos.

D. non-analytic expansion of the ReLU correlation map near $c = 1$

In this appendix we derive the non-analytic expansion

$$F_{\text{ReLU}}(1 - m) = 1 - m + \kappa m^{3/2} + \mathcal{O}(m^2), \quad \kappa = \frac{2\sqrt{2}}{3\pi}, \quad (97)$$

starting from the closed form ReLU correlation map

$$F_{\text{ReLU}}(c) = \frac{1}{\pi} \left[\sqrt{1 - c^2} + (\pi - \cos^{-1} c) c \right], \quad c \in [-1, 1]. \quad (98)$$

Set $c = 1 - m$ with $0 < m \ll 1$.

First expand the square root term,

$$\begin{aligned} \sqrt{1 - c^2} &= \sqrt{1 - (1 - m)^2} = \sqrt{2m - m^2} = \sqrt{2m} \sqrt{1 - \frac{m}{2}} \\ &= \sqrt{2m} \left(1 - \frac{m}{4} + \mathcal{O}(m^2) \right) = \sqrt{2} m^{1/2} - \frac{\sqrt{2}}{4} m^{3/2} + \mathcal{O}(m^{5/2}). \end{aligned} \quad (99)$$

Next expand $\cos^{-1}(1 - m)$. Write $\theta = \cos^{-1}(1 - m)$ and use

$$\cos \theta = 1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} + \mathcal{O}(\theta^6), \quad (100)$$

together with an ansatz $\theta = a m^{1/2} + b m^{3/2} + \mathcal{O}(m^{5/2})$. Matching $\cos \theta = 1 - m$ order by order gives $a = \sqrt{2}$ and $b = \sqrt{2}/12$, hence

$$\cos^{-1}(1 - m) = \sqrt{2} m^{1/2} + \frac{\sqrt{2}}{12} m^{3/2} + \mathcal{O}(m^{5/2}). \quad (101)$$

Now expand the product term in (98),

$$\begin{aligned} (\pi - \cos^{-1} c) c &= \left(\pi - \cos^{-1}(1 - m) \right) (1 - m) \\ &= \left(\pi - \sqrt{2} m^{1/2} - \frac{\sqrt{2}}{12} m^{3/2} + \mathcal{O}(m^{5/2}) \right) (1 - m) \\ &= \pi(1 - m) - \sqrt{2} m^{1/2} + \frac{11\sqrt{2}}{12} m^{3/2} + \mathcal{O}(m^2). \end{aligned} \quad (102)$$

Adding (99) and (102) cancels the $\mathcal{O}(m^{1/2})$ terms and yields

$$\begin{aligned} \sqrt{1 - c^2} + (\pi - \cos^{-1} c) c &= \pi(1 - m) + \left(-\frac{\sqrt{2}}{4} + \frac{11\sqrt{2}}{12} \right) m^{3/2} + \mathcal{O}(m^2) \\ &= \pi(1 - m) + \frac{2\sqrt{2}}{3} m^{3/2} + \mathcal{O}(m^2). \end{aligned} \quad (103)$$

Dividing by π gives (97) with $\kappa = 2\sqrt{2}/(3\pi)$.

The key structural point is the cancellation of the $\mathcal{O}(m^{1/2})$ pieces, leaving a leading non-analytic correction of order $m^{3/2}$, which encodes the kink.

E. Square root tube origin of the $m^{3/2}$ non-analyticity for kinked activations

This appendix explains why the near alignment expansion of the correlation map develops a universal fractional power for kinked activations. The main text derives the ReLU result in closed form and finds

$$F(1 - m) = 1 - m + \kappa m^{3/2} + \mathcal{O}(m^2), \quad (104)$$

with $\kappa = \frac{2\sqrt{2}}{3\pi}$ for ReLU. Here we justify why the exponent $3/2$ is generic for any ordinary kink, while the amplitude is nonuniversal.

Let (u_1, u_2) be jointly Gaussian with

$$\mathbb{E}[u_1] = \mathbb{E}[u_2] = 0, \quad \mathbb{E}[u_1^2] = \mathbb{E}[u_2^2] = 1, \quad \mathbb{E}[u_1 u_2] = c. \quad (105)$$

The normalized correlation map takes the form

$$F(c) \propto \mathbb{E}[\phi(u_1)\phi(u_2)], \quad (106)$$

with proportionality fixed by the variance normalization used in the main text.

We study the approach to perfect alignment by writing

$$c = 1 - m, \quad 0 < m \ll 1. \quad (107)$$

Introduce the sum and difference coordinates

$$s = \frac{u_1 + u_2}{\sqrt{2}}, \quad d = \frac{u_1 - u_2}{\sqrt{2}}. \quad (108)$$

A direct covariance computation gives

$$\text{Var}(s) = 1 + c = 2 - m, \quad \text{Var}(d) = 1 - c = m, \quad \mathbb{E}[sd] = 0. \quad (109)$$

Hence as $m \rightarrow 0$ one has the scaling

$$s \sim O(1), \quad d \sim O(\sqrt{m}). \quad (110)$$

Geometrically, the joint Gaussian measure concentrates in a thin tube of transverse thickness $\sqrt{m}$ around the diagonal $u_1 = u_2$.

As we now show, this is not an idiosyncrasy of ReLU, but rather a general feature of kinked activations. The key point is that, in a neighborhood of the kink, any ordinary kinked ϕ can be written as a smooth part plus a multiple of ReLU. Concretely, if ϕ has finite one-sided slopes $a_{\pm}$ at $u = 0$, then locally one may decompose

$$\phi(u) = a_- u + (a_+ - a_-) \text{ReLU}(u) + r(u), \quad r(u) = o(u) \text{ as } u \rightarrow 0, \quad (111)$$

where r is continuous and has no slope jump at $u = 0$. The remainder r contributes only integer-power terms in m in the near-alignment expansion, so the leading fractional-power term is controlled by the ReLU component and therefore has exponent $3/2$ for every ordinary kink; only its amplitude depends on nonuniversal details such as $a_{\pm}$ and the precise normalization.

F. Variational approach to optimal activations

In this appendix, we show that the family of functions that are most natural for our variational problem with Gaussian measure are Hermite polynomials, and decomposing activation functions into Hermite polynomials can give us a simple and effective tool to analyze the behavior of activations on this basis. Concentrating spectral weight in low Hermite degrees makes correlations more stable with depth (smaller χ) but yields nearly linear features, whereas populating higher degrees increases non-linearity at the cost of larger susceptibilities and hence shorter depth scales unless the initialization is retuned.

We also address the scale redundancy $\phi \mapsto a \phi$, which can be compensated by $\sigma_w^2 \mapsto \sigma_w^2/a^2$ without changing the mean-field fixed point. To isolate shape dependence, we fix a variance fixed point q^* and the bias variance σ_b^2 , and treat σ_w^2 as determined by the fixed-point condition. With this convention, the remaining freedom is the shape of a function $f \in L^2(Dz)$ with Gaussian inner product

$$Dz \equiv \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz, \quad \langle f, g \rangle \equiv \int Dz f(z) g(z).$$

F.1. A scale-fixed variational problem for χ

We define the fixed-point rescaled activation

$$f(z) \equiv \phi(\sqrt{q^*} z).$$

Assuming ϕ is weakly differentiable with $f \in H^1(Dz)$, Price's theorem yields the correlation susceptibility (Schoenholz et al., 2017)

$$\chi \equiv F'(1) = \frac{\sigma_w^2}{q^*} \int Dz (f'(z))^2. \quad (112)$$

The variance fixed-point equation

$$q^* = \sigma_w^2 \int Dz f(z)^2 + \sigma_b^2$$

determines

$$\sigma_w^2 = \frac{q^* - \sigma_b^2}{\int Dz f^2}.$$

Substituting gives the scale-fixed factorization

$$\chi = \left(1 - \frac{\sigma_b^2}{q^*}\right) \frac{\int Dz (f')^2}{\int Dz f^2} \equiv \left(1 - \frac{\sigma_b^2}{q^*}\right) \mathcal{Q}[f], \quad \mathcal{Q}[f] \equiv \frac{\|f'\|_2^2}{\|f\|_2^2}. \quad (113)$$

Thus, at fixed (q^*, σ_b^2) , extremizing χ over activations reduces to extremizing the Rayleigh quotient (see for example (Horn & Johnson, 1985)) $\mathcal{Q}[f]$ over the shape of f . Criticality $\chi = 1$ is equivalent to the constraint

$$\mathcal{Q}[f] = \left(1 - \frac{\sigma_b^2}{q^*}\right)^{-1}.$$

In this convention, $\mathcal{Q}[f]$ quantifies the activation's effective sensitivity to typical Gaussian fluctuations: shifting spectral weight to higher modes increases $\|f'\|_2$ relative to $\|f\|_2$ and therefore increases χ unless σ_w^2 is retuned.

F.2. The Ornstein–Uhlenbeck eigenvalue problem

Because $\mathcal{Q}[f]$ is scale-invariant, we may impose the normalization $\int Dz f^2 = 1$ and extremize $\int Dz (f')^2$. Introducing a Lagrange multiplier λ , consider

$$\mathcal{S}[f] \equiv \int Dz \left[(f'(z))^2 - \lambda f(z)^2 \right].$$

Gaussian-weighted integration by parts (with boundary terms vanishing under mild decay assumptions) gives the Euler–Lagrange equation

$$(-\partial_z^2 + z \partial_z) f(z) = \lambda f(z), \quad (114)$$

i.e. the Ornstein–Uhlenbeck eigenproblem on $L^2(Dz)$. Its spectrum is

$$\lambda_n = n, \quad n = 0, 1, 2, \dots,$$

and the eigenfunctions form a complete orthonormal basis. Under the unit-norm constraint, these eigenfunctions are stationary points of $\mathcal{Q}[f]$, with $\mathcal{Q}[f] = \lambda$.

F.3. Hermite basis and diagonal correlation propagation

A natural orthonormal eigenbasis of (114) is given by the normalized Hermite polynomials

$$h_n(z) \equiv \frac{1}{\sqrt{2^n n!}} H_n\left(\frac{z}{\sqrt{2}}\right), \quad \langle h_n, h_m \rangle = \delta_{nm},$$

where H_n are the physicists' Hermite polynomials.⁴ They satisfy

$$h'_n(z) = \sqrt{n} h_{n-1}(z), \quad (-\partial_z^2 + z \partial_z) h_n(z) = n h_n(z), \quad \mathcal{Q}[h_n] = n.$$

Moreover, this basis diagonalizes correlated-Gaussian expectations: if (z_1, z_2) are jointly Gaussian with unit variances and correlation $c \in [-1, 1]$, then

$$\mathbb{E}[h_n(z_1) h_m(z_2)] = c^n \delta_{nm}. \quad (115)$$

Thus each Hermite degree propagates independently through the Gaussian channel with attenuation c^n .

Expanding an arbitrary $f \in L^2(Dz)$ in this basis,

$$f(z) = \sum_{n \geq 0} a_n h_n(z), \quad a_n = \langle f, h_n \rangle, \quad \sum_{n \geq 0} a_n^2 = \int Dz f^2,$$

we obtain

$$\mathbb{E}[f(z_1) f(z_2)] = \sum_{n \geq 0} a_n^2 c^n.$$

Consequently, the correlation map takes the form

$$F(c) = \frac{\sigma_w^2 \sum_{n \geq 0} a_n^2 c^n + \sigma_b^2}{q^*}, \quad q^* = \sigma_w^2 \sum_{n \geq 0} a_n^2 + \sigma_b^2.$$

Analyticity of $F(c)$ near $c = 1$ is controlled by the decay of $|a_n|$: rapid decay (typical for smooth activations) yields an ordinary Taylor expansion around $c = 1$, while slow decay (typical for kinked or piecewise smooth activations) allows infinitely many high-degree modes to contribute collectively and can produce non-Taylor behavior (e.g. fractional powers or logarithms) as $c \rightarrow 1$, corresponding to a branch point at $c = 1$.

Using $h'_n = \sqrt{n} h_{n-1}$, we can compute the susceptibility

$$\chi = F'(1) = \frac{\sigma_w^2}{q^*} \sum_{n \geq 1} n a_n^2. \quad (116)$$

⁴Equivalently, $h_n(z) = \text{He}_n(z)/\sqrt{n!}$ in terms of the probabilists' Hermite polynomials He_n .

In the zero-bias case $\sigma_b^2 = 0$ this becomes

$$\chi = \frac{\sum_{n \geq 1} n a_n^2}{\sum_{n \geq 0} a_n^2} = \mathcal{Q}[f],$$

so χ is the mean Hermite degree under weights proportional to a_n^2 , and criticality $\chi = 1$ corresponds to unit mean degree.

For completeness, the variance-channel susceptibility $\chi_q \equiv \partial q^{(\ell+1)} / \partial q^{(\ell)}|_{q^*}$ can be written in the same basis as

$$\chi_q = \frac{\sigma_w^2}{q^*} \left(\sum_{n \geq 1} n a_n^2 + \sum_{n \geq 0} \sqrt{(n+1)(n+2)} a_n a_{n+2} \right). \quad (117)$$

Thus χ depends only on the distribution $\{a_n^2\}$ across Hermite degrees, whereas χ_q is additionally sensitive to relative signs/phases within each parity subsector through the $a_n a_{n+2}$ couplings.

Pure modes $f \propto h_n$ are stationary points with $\mathcal{Q}[h_n] = n$. It is therefore easy to see that the $n = 1$ mode is going to be the global minimum, though this reduces our MLP to a linear network. Any genuine non-linearity necessarily populates higher degrees and increases $\mathcal{Q}[f]$, shortening the correlation depth scale unless σ_w^2 is reduced accordingly. The Hermite coefficients $\{a_n\}$ therefore provide a concrete parametrization of the expressivity–trainability tradeoff.

F.4. Smooth and kinked activations

For smooth activations, the Hermite coefficients typically decay rapidly and $F(c)$ is analytic near $c = 1$, with χ dominated by the lowest few degrees. For kinked activations such as ReLU, $\phi'(u) = \theta(u)$ almost everywhere, so

$$\chi = \sigma_w^2 \int Dz \theta(u) = \frac{\sigma_w^2}{2}, \quad \chi = 1 \Rightarrow \sigma_w^2 = 2.$$

In our scale-fixed convention (113), ReLU has $\mathcal{Q}[f] = 1$ (for $f(z) \propto \max(0, z)$), and hence $\chi = (1 - \sigma_b^2/q^*)$; achieving $\chi = 1$ in that convention forces $\sigma_b^2 = 0$. In Hermite space, the kink generates a slow (power-law) tail in $|a_n|$, and the resulting series for $F(c)$ develops a branch point as $c \rightarrow 1$; the non-analyticity originates from the collective contribution of high-degree modes.

n	$h_n(z)$ parity	a_n for ReLU(z)	a_n for tanh(z)
0	even	$\frac{1}{\sqrt{2\pi}}$	0
1	odd	$\frac{1}{2}$	0.60570551
2	even	$\frac{1}{2\sqrt{\pi}}$	0
3	odd	0	-0.14843719
4	even	$-\frac{1}{\sqrt{48\pi}}$	0
5	odd	0	0.06254752
6	even	$\frac{1}{4\sqrt{10\pi}}$	0
7	odd	0	-0.03144542
8	even	$-\frac{15}{\sqrt{80640\pi}}$	0
9	odd	0	0.01741993
10	even	$\frac{105}{\sqrt{7257600\pi}}$	0
11	odd	0	-0.01029184

Table 11. Orthonormal Hermite coefficients $a_n = \int Dz f(z) h_n(z)$ for two standard activations, shown for the unscaled choices $f(z) = \text{ReLU}(z)$ and $f(z) = \tanh(z)$. For $f(z) = \tanh(sz)$, use $a_n(s) = \int Dz \tanh(sz) h_n(z)$, with $a_{2k}(s) = 0$ by odd parity.

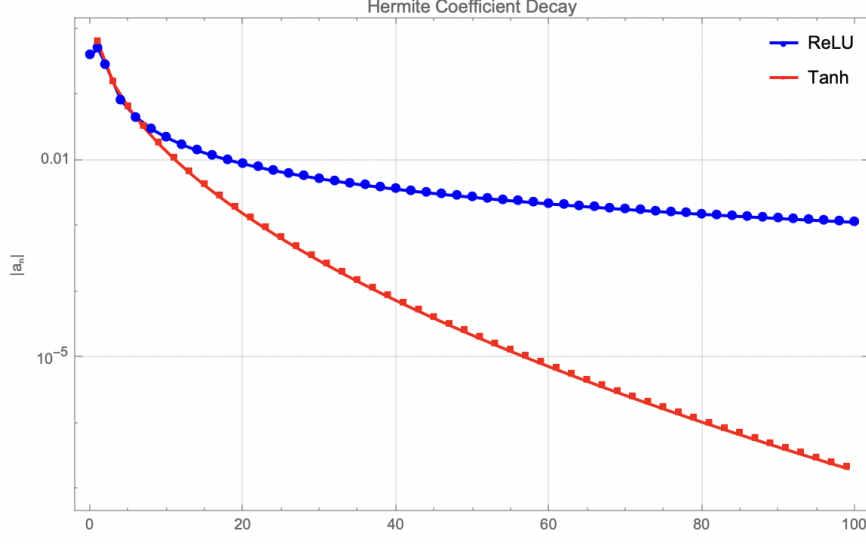


Figure 10. Magnitude of Hermite coefficients $|a_n|$ for ReLU versus tanh. ReLU exhibits a slow (power-law) decay, reflecting multi-scale support across Hermite degrees, while tanh decays rapidly and concentrates most spectral mass in the lowest modes.

F.5. Hermite decompositions for ReLU and tanh

Throughout, $Z \sim \mathcal{N}(0, 1)$ and

$$Dz \equiv \frac{dz}{\sqrt{2\pi}} e^{-z^2/2}. \quad (118)$$

We expand the fixed point rescaled activation $f(z)$ in an orthonormal Hermite basis of $L^2(Dz)$,

$$f(z) = \sum_{n \geq 0} a_n h_n(z), \quad a_n = \int Dz f(z) h_n(z), \quad \int Dz h_n(z) h_m(z) = \delta_{nm}. \quad (119)$$

With physicists Hermite H_n ,

$$H_n(x) \equiv (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}, \quad h_n(z) \equiv \frac{1}{\sqrt{2^n n!}} H_n\left(\frac{z}{\sqrt{2}}\right). \quad (120)$$

F.5.1. ReLU

Let $f(z) = \text{ReLU}(z) = \max(0, z)$. The coefficients have a closed form.

$$a_0 = \frac{1}{\sqrt{2\pi}}, \quad a_1 = \frac{1}{2}, \quad a_{2k+1} = 0 \text{ for } k \geq 1, \quad (121)$$

and for $k \geq 1$,

$$a_{2k} = \frac{(-1)^{k-1} (2k-3)!!}{\sqrt{2\pi} (2k)!}. \quad (122)$$

We use the convention $(-1)!! = 1$. If instead $f(z) = \text{ReLU}(\sqrt{q^*} z)$, then all coefficients scale as $a_n \mapsto \sqrt{q^*} a_n$.

F.5.2. tanh

Let $f(z) = \tanh(sz)$ with $s > 0$. The coefficients are

$$a_n(s) = \int Dz \tanh(sz) h_n(z). \quad (123)$$

Parity gives $a_{2k}(s) = 0$. There is no simple closed form for general s , but the coefficients are easy to compute numerically by Gaussian quadrature using Mathematica, or other specialized packages.